

L-functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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Abstract

We give a single manifestly finite, regulator-free functional realizing in one object L-functions quasi-periods and period polynomials of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with generic poles throughout the upper half-plane $\mathbb{H}$ and at the cusp of $\mathrm{SL}_2(\mathbb{Z})$. Additionally, our construction preserves holomorphicity. It is a two-segment contour integral, $L^*(f, s) = L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S)$, where L_T^* and L_S^* have distinct integration kernels, and $\gamma_{\tau_0}^S$ is a path in $\mathbb{H}$ from $S(\tau_0)$ to τ_0 , and the same for $\gamma_{\tau_0}^T$. For weakly holomorphic forms our L-function is independent of τ_0 and of the path, and reproduces classical L-functions for holomorphic modular forms, reproduces the L-functions of weak modular forms in [1], which previously needed non-holomorphic regularization to be defined, and reproduces Brown's quasi-periods [2]. For properly meromorphic modular forms with poles inside the interior of $\mathbb{H}$, our L-function depends on the homotopy classes, defined in terms of pole-crossings. There is a natural choice of trivial homotopy class, which renders our L-functions canonical when the modular form has a pole in the interior of the fundamental domain, generalizing [3].

1 Introduction and preliminary concepts

Our primary goal in this note is to put forth a unifying generalization of the L-function associated to a given modular form. Our L-function naturally encompasses the classical case of L-functions associated to holomorphic modular forms [4], and seamlessly unifies Ref. [1]'s extension to L-functions of modular forms with poles at the cusp with Ref. [3]'s extension to L-functions of modular forms with poles at finite locations, exclusively away from cusps. Secondly, we show that our L-functions reproduce the numerical results for quasi-periods associated to weak cusp forms [2].

Note: Throughout this document, we refer to L-integrals and L-functions. The classical L-function-as-Dirichlet-series which has been suitably extended to the complex s plane is simply not defined when the modular form that defines the L-function has poles either at the cusp or at finite locations within the upper half-plane. So, if there is ever a doubt, we state clearly: *any L-function in this note is defined by an L-integral*. The focus of this note is to define L-integrals for any integer weight $\mathrm{SL}_2(\mathbb{Z})$ modular form which may have poles at cusps or away from cusps.

1.1 Modular forms

Write $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ for the real and imaginary parts of a complex number $z \in \mathbb{C}$. Let $\mathbb{H} := \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$ denote the upper half-plane. $\mathrm{SL}_2(\mathbb{Z})$ is the group generated by $T := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

There is a natural $\mathrm{SL}_2(\mathbb{Z})$ action on functions $f : \mathbb{H} \rightarrow \mathbb{C}$, graded by integers k : $(f|_k \gamma)(\tau) := (c\tau + d)^{-k} f(\gamma \cdot \tau)$ for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Similarly, $\gamma \cdot \tau := \frac{a\tau + b}{c\tau + d}$.

In this vein, a *holomorphic modular form* of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ such that $(f|_k \gamma)(\tau) = f(\tau)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ (aka $f(\tau + 1) = f(-\frac{1}{\tau})\tau^{-k} = f(\tau)$) for all

$\tau \in \mathbb{H}$, and $f(\tau) = \mathcal{O}(\tau^k)$ as $\text{Im}(\tau) \rightarrow \infty$, and which has no poles at finite locations within $\mathbb{H}$. By virtue of the periodicity $f(\tau) = f(\tau + 1)$ and holomorphicity, each holomorphic modular form has a q -series decomposition,

$$f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau} = \sum_n c_f(n) q^n \quad , \quad q := e^{2\pi i \tau} \quad . \quad (1)$$

In order for f to be bounded as $\tau \rightarrow i\infty$, we conclude that $c_f(n) = 0$ for all $n < 0$, i.e. that $f = \sum_{n \geq 0} c_f(n) q^n$; for such modular forms f we write $f \in M_k$. If a modular form f additionally has a vanishing constant term, i.e. if $c_f(0) = 0$, then f is a *cusp form*, and we write $f \in S_k$. Clearly, $S_k \subset M_k$. Further, M_k is a finite-dimensional vector-space, as is S_k . Holomorphic $\text{SL}_2(\mathbb{Z})$ modular forms must have even integer weights, and $k \geq 4$ [4].

A product of modular forms of weights k_1 and k_2 is a modular form of weight $k_1 + k_2$, so $M_* := \bigoplus_k M_k$ is naturally a (graded) ring. This graded ring is freely generated by the two modular forms $E_4 \in M_4$ and $E_6 \in M_6$, the $k = 4, 6$ specializations of the *weight k Eisenstein series* E_k , which may be defined as follows:

$$E_k := \frac{1}{2} \sum_{(m,n)=1} \frac{1}{(m+n\tau)^k} = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \quad , \quad (2)$$

where $\sum_{(m,n)=1}$ denotes summing over all relatively prime $(m, n) \in \mathbb{Z}^2$, the Bernoulli number B_n is the coefficient of $t^n/n!$ of the Laurent expansion of $t/(e^t - 1)$ about $t = 0$, and $\sigma_m(n) := \sum_{d|n} d^m$ is the Divisor function where $d|n$ means $n/d \in \mathbb{N}$. In short, $M_* = \mathbb{C}[E_4, E_6]$. We now define the *modular discriminant*:

$$\Delta := \frac{E_4^3 - E_6^2}{1728} = q + O(q^2) \in S_{12} \quad . \quad (3)$$

Holomorphicity is most canonically relaxed via allowing a modular form to have a pole at the cusp of $\text{SL}_2(\mathbb{Z})$, i.e. for a weight k modular form for $\text{SL}_2(\mathbb{Z})$ to have negative powers of q in its q -series expansion. In this case, f diverges as $\tau \rightarrow i\infty$ as $f \sim e^{-2\pi N \tau}$, where $-N$ is the largest negative number such that $c_f(n)$ is non-zero. Such a modular form f is called a *weak modular form*, and we write $f \in M_k^!$. If a weak modular form f has a vanishing coefficient of q^0 , i.e. if $c_f(0) = 0$, it is called a *weak cusp form* and we write $f \in S_k^!$. Clearly, $S_k^! \subset M_k^!$, $S_k \subset S_k^!$ and $M_k \subset M_k^!$. Two important weak cusp forms in this note are as follows:

$$J := \frac{E_4^3}{\Delta} - 744 = \frac{1}{q} + O(q) \in S_0^! \quad , \quad (4)$$

$$\widehat{\Delta} = (J^2 + 24J - 393444)\Delta := \frac{1}{q} + O(q^2) \in S_{12}^! \quad . \quad (5)$$

In this work, we further relax from even weakly holomorphic modular forms, and consider *meromorphic modular forms* of weight k which are quotients: $f = \frac{f_1}{f_2}$ where $f_i \in M_{k_i}$, the denominator f_2 is not identically zero, and $k = k_1 - k_2$. We write F_k for the vector space of meromorphic modular forms of weight k , and set $F_* := \bigoplus_k F_k$. Then F_k vanishes unless k is an even integer, and F_* may be regarded as a “graded field of fractions” of M_* .

1.2 L-functions for holomorphic modular forms and periods for cusp forms

For any $f \in S_k$, one may define the associated L-function:

$$L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s) \stackrel{f \in S_k}{:=} \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty f(it) t^{s-1} dt = \sum_{n \geq 1} \frac{c_f(n)}{n^s} \quad . \quad (6)$$

A similar definition applies for $f \in M_k$. (The classical definition of $L^*(f, s)$ for $f \in M_k$ is the focus of Remark 3.6, below.) Similarly, one may define a period polynomial for $f \in S_k$ in terms of this L-function,

$$r_f(X, Y) := (2\pi i)^{k-1} \sum_{\ell=0}^{k-2} (-1)^\ell \binom{k-2}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{k-2-\ell} Y^\ell. \quad (7)$$

This polynomial, $r_f(X, Y) \in V_{k-2} := \mathbb{C}[X, Y]_{k-2}$, has a natural action under $\mathrm{SL}_2(\mathbb{Z})$, $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$ for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. This period polynomial can be decomposed into a basis of highly constrained polynomials, which are annihilated by $|(1 + S)$ and $|(1 + U + U^2)[6]$, where $U := TS$ generates an order-3 cyclic subgroup of $\mathrm{SL}_2(\mathbb{Z})$. We define the *periods* of a cusp form to be coefficients of r_f within this highly constrained sub-space W of V_{k-2} , i.e. $W \subset V_{k-2}$.

Why this work? As classically computed, these L-functions and associated period polynomials (when defined in terms of the special values of the associated L-function) are not defined when f has poles either at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ or within the interior of $\mathbb{H}$. As stated above, there are extant, but partial and non-overlapping, extensions of the definition of $L(f, s)$ to $f \in M_k^!$ [1] or $f \in F_k$ when f is regular at the cusp of $\mathrm{SL}_2(\mathbb{Z})$ [3]. In this note, we provide a construction that unifies these disparate extensions. We start with generic L-integrals in section 2, then specialize to $f \in M_k^!$ in section 3, and finally explore the more general context of $f \in F_k$ in section A.

2 A finite L-integral

Definition 2.1. Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ denote the generators of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $f^{-1}(\infty)$ be the locus of all poles of $f \in \mathbb{H}$ (inclusive of the cusp of $\mathrm{SL}_2(\mathbb{Z})$). The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ from $\sigma^{-1}\tau_0$ to τ_0 , for $\sigma = S$ and $\sigma = T$ respectively,

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0. \quad (8)$$

Avoiding poles is automatic for $f \in M_k^!$, as long as the base-point τ_0 , is not at a cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. there is only the trivial homotopy class. When $f \in F_k$, and f has poles at finite and non-zero imaginary heights, integration contours fall into nontrivial homotopy classes.

Definition 2.2. Let $\zeta(s, x) = \sum_{m \geq 0} (m + x)^{-s}$ as a function $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \mathbb{N})_x \rightarrow \mathbb{C}$ denote the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$). We define Hurwitz T -kernel as

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(1 - (k - s), \tau + 1). \quad (9)$$

Definition 2.3 (Generic L-integral). Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$. Further, let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 2.1, and let $\tilde{k}_T$ be as defined in Definition 2.2. The L-integral of f is

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left(\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right) \quad (10)$$

We use this L -integral to define a generalized L -function:

Definition 2.4 (Generic L-function). Let $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ be as defined in Definition 2.3. We define the generalized L-function of $f \in F_k$ to be,

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (11)$$

Having defined the L-integral and L-function, we prove they are manifestly finite for $f \in F_k$:

Lemma 2.5 (Finiteness of L-integrals). *For $f \in F_k$, both $L_S^*(f, s; \gamma_{\tau_0}^S)$ and $L_T^*(f, s; \gamma_{\tau_0}^T)$ in Definition 2.3 are each, separately finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and for every path $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau + 1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 2.2). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

We now turn to path-dependence and τ_0 -independence of the L-integral and the L-function of $f \in F_k$. This holds for all $f \in M_k^!$:

Proposition 2.6 (Basepoint & path independence for $f \in M_k^!$). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 show that variations in τ_0 for the S-term and T-term in $L^*(f, s)$ cancel, i.e. that $\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = -\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T)$. For L_T^* , note that the T -segment runs from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 , so by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (12)$$

where $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$. This is the whole role of δ_T : the unit gap between the endpoints turns the ∂_{τ_0} boundary term into a unit shift of the kernel. That shift is elementary, since $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$ gives, on both terms at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)} (\tau + 1)^{k-1-s} \quad (13)$$

$$\implies \partial_{\tau_0} L_T^*(f, s; \gamma_{\tau_0}^T) = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (14)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$,

$$\partial_{\tau_0} L_S^*(f, s; \gamma_{\tau_0}^S) = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s}]. \quad (15)$$

Thus τ_0 -variations are equal and opposite, $\partial_{\tau_0} L_S^* = -\partial_{\tau_0} L_T^*$. Thus $\partial_{\tau_0} L^* = \partial_{\tau_0} L_S^* + \partial_{\tau_0} L_T^* = 0$. $\square$

Remark 2.7 (Recovering the classical L-integral for $f \in S_k$). Let $f \in S_k$. As $f^{-1}(\infty)$ does not include the cusp $i\infty$, $\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ exists. In this limit, the S -segment maps onto the standard L-function contour, $[0, i\infty]$. Simultaneously, as $f(it) \sim e^{-2\pi t}$ as $t \rightarrow \infty$, the T -segment vanishes exponentially. Thus, in this limit, $L^*(f, s)$ exactly maps onto the classical L-function integral [4]:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt. \quad (16)$$

3 L-functions, period polynomials and poles at the cusp

Proposition 3.1. *Let $\Gamma(s, x) := \int_x^\infty e^{-t} t^{s-1} dt$ denote the incomplete Gamma function. For $n \neq 0$, $\nu \in \mathbb{C}$, and $z \in \mathbb{H}$,*

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (17)$$

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one horizontal ray,

$$\int_z^{z+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{z+m}^{z+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (18)$$

Now, substitute $w = -2\pi i n v$, so that $v = iw/(2\pi n)$, $dv = i dw/(2\pi n)$, and $e^{2\pi i n v} = e^{-w}$: the lower endpoint becomes $w = -2\pi i n z$, and for $n > 0$ the ray rotates to $w \rightarrow +\infty$ along the positive reals. Since $e^{-w} w^{-\nu}$ decays in the right half-plane, the closing arc vanishes. With $v^{-\nu} = i^{-\nu} (2\pi n)^{\nu} w^{-\nu}$,

$$\int_z^{z+\infty} e^{2\pi i n v} v^{-\nu} dv = i^{1-\nu} (2\pi n)^{\nu-1} \int_{-2\pi i n z}^{\infty} e^{-w} w^{-\nu} dw = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n z). \quad (19)$$

For $n < 0$ the ray instead grows as $\text{Im} v \rightarrow \infty$ and the rotation is illegal; but both sides are analytic in n and agree for $n > 0$, so the identity continues to $n < 0$ with Γ on its principal branch. At $n = 0$ the prefactor $(2\pi n)^{\nu-1}$ is singular; that case is handled separately. $\square$

Theorem 3.2 (The L-function for $f \in M_k^!$). *For $f \in M_k^!$, we may set $\tau_0 = i$, and prove*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right). \quad (20)$$

Here, trivially, $L^*(f, s) = i^k L^*(f, k-s)$. The L-function is $L(f, s) := (2\pi)^s L^*(f, s)/\Gamma(s)$.

Proof. By path independence and base-point independence from Proposition 2.6, we may freely set $\tau_0 = i$. Here, $L_S^*(f, s; \gamma_{\tau_0}^S) = 0$ (the S -contour vanishes in length and there are no nontrivial homotopy classes as $f \in M_k^!$ has no poles which a finite contour can wrap), and the L-integral simplifies dramatically:

$$L^*(f, s) = L_T^*(f, s; \gamma_i^T) = e^{-\pi i s/2} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

We then expand f in terms of its q-series, $\sum_n c_f(n) e^{2\pi i n \tau}$, and substitute the u -variable of Proposition 3.1 as $u = \tau + 1$, and note that $e^{2\pi i n(u-1)} = e^{2\pi i n u}$. When $n \neq 0$, straightforward substitutions of ν, z into Proposition 3.1 suffice: $1-\nu = s$ gives the contribution of the first term in $\tilde{k}_T(\tau, s)$, and $1-\nu = k-s$ gives the second:

$$e^{-\pi i s/2} \left(e^{\pi i s/2} \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} - e^{i\pi(s-1)} e^{\pi i(k-s)/2} \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (22)$$

We may then evaluate the $n = 0$ mode's contribution follows from noting the closed form solution for the definite integral,

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (23)$$

This result follows from noting $\partial_a(\zeta(\sigma-1, a)/(1-\sigma)) = \zeta(\sigma, a)$ and $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$. Applying this to the two distinct terms of $\tilde{k}_T(\tau, s)$ at $\sigma = 1-s$ and $\sigma = 1-(k-s)$ gives

$$e^{-\pi is/2} \int_{i-1}^i c_f(0) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (24)$$

Combining the $n=0$ contributions with the $n \neq 0$ contributions reproduces Eq. (20). Finally, the functional equation, $L^*(f, s) = i^k L^*(f, k-s)$, follows by direct inspection. $\square$

3.1 Explicit S- and T-contours for general τ_0

Lemma 3.3 (L_T^* for $M_k^!$). For $f \in M_k^!$, and $\tau_0 \in \mathbb{H}$ away from cusps, $L_T^*(f, s; \gamma_{\tau_0}^T)$ evaluates to:

$$L_T^*(f, s; \gamma_{\tau_0}^T) = -c_f(0) \left(\frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (25)$$

Proof. Expand $f = \sum_n c_f(n) q^n$ on the compact segment $[\tau_0 - 1, \tau_0]$, and restrict attention to the sub-sum, $\sum_{n \neq 0}$. For each such $n \neq 0$, apply the Proposition 3.1 in $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$ at $\nu = 1-s$ and $\nu = s-(k-1)$. The phases for the second term work-out identically as in the proof of Theorem 3.2.

For $n=0$, the constant $c_f(0)$ multiplies $e^{-\pi is/2} \int_{\tau_0-1}^{\tau_0} \tilde{k}_T(\tau, s) d\tau$, which we compute as follows. First, the closed-form of the Hurwitz antiderivative $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1)$ comes from $\partial_\tau \zeta(\sigma-1, \tau+1) = -(\sigma-1)\zeta(\sigma, \tau+1)$ and the shift identity $\zeta(\sigma-1, \tau_0+1) - \zeta(\sigma-1, \tau_0) = -\tau_0^{1-\sigma}$. Finally, note that the two terms in $\tilde{k}_T$ are simply specializations of $\sigma = 1-s$ and $\sigma = s-k+1$, which yield $-e^{-\pi is/2} \tau_0^s/s = -(\tau_0/i)^s/s$ and $-e^{\pi is/2} \tau_0^{k-s}/(k-s) = -i^k (\tau_0/i)^{k-s}/(k-s)$ respectively. $\square$

Corollary 3.4 (L_T^* as $\tau_0 \rightarrow i\infty$). Let $f \in S_k^!$, where $f = \sum_{n \geq -N} c_f(n) q^n$ for some $-N \in \mathbb{Z}$. The T-contour's contribution to $L^*(f, s)$ is given by Lemma 3.3. As $\tau_0 = it$ goes to the cusp $i\infty$, the T-contour scales exactly as $f(it)$ scales: $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(it) \sim e^{2\pi Nt}$.

Proof. For the asymptotics, $\Gamma(\sigma, z) \sim z^{\sigma-1} e^{-z}$ as $|z| \rightarrow \infty$ for $|\arg z| < \frac{3\pi}{2}$ [10, §8.11] gives $|c_f(n)|$ -weighted go as $e^{-2\pi n \text{Im}(\tau_0)}$. The smallest mode n in the q-series, i.e. $n = -N$, dominates the scaling of $f(it)$ and of $L_T^*(f, s; \gamma_{it}^T)$. Combining these results yields the claimed asymptotics. $\square$

Now, exploiting τ_0 -independence of $L^*(f, s)$, its explicit value as given in Theorem 3.2, and the explicit T-contribution $L_T^*(f, s; \gamma_{\tau_0}^T)$ trivially implies clean, closed-form expressions for $L_S^*(f, s; \gamma_{\tau_0}^S)$:

Corollary 3.5 (L_S^* for $M_k^!$). For $f \in M_k^!$ and $\tau_0 \in \mathbb{H}$, $L_S^*(f, s; \gamma_{\tau_0}^S)$ evaluates to

$$L_S^*(f, s; \gamma_{\tau_0}^S) = c_f(0) \left(\frac{(\tau_0/i)^s - 1}{s} + i^k \frac{(\tau_0/i)^{k-s} - 1}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left(\frac{\Gamma(s, 2\pi n) - \Gamma(s, -2\pi in\tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n) - \Gamma(k-s, -2\pi in\tau_0)}{(2\pi n)^{k-s}} \right). \quad (26)$$

Proof. Recall $L^*(f, s) - L_T^*(f, s; \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S)$. Now, $L^*(f, s)$ for $f \in M_k^!$ is given by Theorem 3.2, and $L_T^*(f, s; \gamma_{\tau_0}^T)$ is given by Lemma 3.3. Their difference yields Eq. (26). $\square$

3.2 Classical Mellin recovery and canonical uniqueness.

As discussed in Remark 2.7, for $f \in S_k$, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in Theorem 3.2 collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while via Corollary 3.4 the T -segment vanishes exponentially as $L_T^*(f, s; \gamma_{\tau_0}^T) \sim f(\tau_0) \sim e^{-2\pi\text{Im}(\tau_0)N}$ (as $f \in S_k \Leftrightarrow f = \sum_{n \geq N} c_f(n)q^n$ with $N \geq 1$). I.e., here, one recovers the classical L-integral of f without obstruction:

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt. \quad (27)$$

This story changes when $f = 1/q^N(1 + O(q)) \in S_k^!$ for some $N \geq 1$.

In short, when f has a polar q-series, it has a pole at the cusp of $\text{SL}_2(\mathbb{Z})$. Thus, the limit $\tau_0 \rightarrow i\infty$, which was crucial to recovering the classical Mellin contour in Remark 2.7 simply does not exist.

However, it can still be instructive to examine $L_T^*(f, s; \gamma_{\tau_0}^T)$, and $L_S^*(f, s; \gamma_{\tau_0}^S)$ as τ_0 goes towards $i\infty$. In this context, while of course the sum $L_T^*(f, s; \gamma_{\tau_0}^T) + L_S^*(f, s; \gamma_{\tau_0}^S) = L^*(f, s)$ is manifestly invariant under changes of the base-point, each of these two terms can indeed change values. Indeed, Corollary 3.4 clearly shows that as $\tau_0 \rightarrow i\infty$, when $f \in S_k^!$ has negative powers of q in its q-series, then $L_T^*(f, s; \gamma_{\tau_0}^T) \sim e^{2\pi N \text{Im}(\tau_0)}$. Discarding $\gamma_{\tau_0}^T$ therefore does violence to the result: When the S -contour is isolated from its T-partner, and $\tau_0 \rightarrow i\infty$, it diverges. This is the precise reason that [1] needed to regularize their divergent L-integral representation for properly weak cusp forms.

That said, there is one further aspect of our L-function even in the classical case of $f \in M_k$ that is worthy of remark:

Remark 3.6 (Recovering the classical L-integral for $f \in M_k$). In textbook treatments [4], when $f \in M_k$, its L-integral is defined as

$$\begin{aligned} L^*(f, s)|_{\text{cl}} &:= \int_0^\infty (f(it) - c_f(0))t^{s-1}dt = \frac{\Gamma(s)}{(2\pi)^s} \sum_{n=1}^\infty \frac{c_f(n)}{n^s} \\ &= -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n=1}^\infty c_f(n) \left(\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right) \end{aligned} \quad (28)$$

Classically, this constant-term subtraction is needed because $\int_0^\infty t^{s-1}dt$ is a pure divergence, so the q-series cannot commute with the integration contour.

However, as is clear from our general Theorem 3.2 for L-integrals for $f \in M_k^!$, and from the specific results of the S- and T-contours in Lemma 3.3 and Corollary 3.5, our L-integral requires no such explicit subtraction of the constant term $-c_f(0)$. Thus, not only does our prescription naturally handle divergences at poles when $f \in M_k^!$ has polar terms in its q-series, but it also obviates the need to by-hand remove the constant mode from the kernel of the L-integral that is needed when defining L-functions in terms of the classical Mellin contour, $\int_0^\infty$.

3.3 Period polynomials, periods and quasi-periods

Because $L^*(f, s)$ is canonical for every $f \in S_k^!$, and because period polynomials for a given modular form can be written in terms of their L-functions evaluated at integer- s in the critical strip, they need no separate construction.

Definition 3.7 (Period polynomial for $S_k^!$). For $f \in S_k^!$ and $n = k - 2$, the period polynomial is

$$r_f(X, Y) := (2\pi i)^{n+1} \sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^\ell. \quad (29)$$

Proposition 3.8 (Period polynomial as a contour integral). *Let $n := k - 2$. With $(X - \tau Y)^n$ in place of the L -weight τ^{s-1} , and $\mathbf{K}_T(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \tilde{k}_T(\tau, \ell + 1) X^{n-\ell} Y^\ell$ in place of $\tilde{k}_T$, the same two segments assemble r_f directly:*

$$r_f(X, Y) = (2\pi i)^{n+1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) (X - \tau Y)^n d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \mathbf{K}_T(\tau; X, Y) d\tau \right]. \quad (30)$$

Proof. By Definition 2.3, the $X^{n-\ell} Y^\ell$ coefficient of the bracket is

$$r_f|_{X^{n-\ell} Y^\ell} = (-1)^\ell \binom{n}{\ell} i^{\ell+1} [L_S^*(f, \ell + 1) + L_T^*(f, \ell + 1)] = (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell + 1). \quad (31)$$

This matches Definition 3.7 term by term, and proves the claim. $\square$

Having now defined the period polynomial associated to a weak cusp form $f \in S_k^!$, we now seek to find its (*quasi*-)periods. First, a number of helpful definitions.

All objects below take values in $V_n := \mathbb{C}[X, Y]_n$, $n = k - 2$, with the slash action $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$ of §1. A map $\gamma \mapsto A_\gamma \in V_n$ is a 1-cocycle for $\mathrm{SL}_2(\mathbb{Z})$ if

$$A_{\gamma\delta} = A_\gamma|_\delta + A_\delta \quad \text{for all } \gamma, \delta \in \mathrm{SL}_2(\mathbb{Z}), \quad (32)$$

and is *parabolic* if in addition $A_T = 0$. For fixed $P \in V_n$ the map $\gamma \mapsto P|_\gamma - P$ is a *coboundary*, and satisfies (32) identically, since $(P|_\gamma - P)|_\delta + (P|_\delta - P) = P|_{\gamma\delta} - P$. Adding or subtracting a coboundary from a cocycle therefore returns a cocycle, which is what lets C_γ below be replaced by a representative $\hat{C}_\gamma$ normalised so that $\hat{C}_T = 0$.

For starters, let $f \in S_k^!$, $\tau_0 \in \mathbb{H}$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, and define

$$C_\gamma(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f, \quad \text{where } \omega_f := f(\tau)(X - \tau Y)^n d\tau. \quad (33)$$

Finally, it is helpful to define two slight abstractions related to $\mathbf{K}_T(\tau; X, Y)$ and to $L_T^*(f, s; \gamma_{\tau_0}^T)$:

$$\chi(\tau; X, Y) := \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell \zeta(-\ell, \tau + 1) X^{n-\ell} Y^\ell, \quad (34)$$

$$E(f; \tau_0) := (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \chi(\tau; X, Y) d\tau. \quad (35)$$

At non-positive integer first argument the Hurwitz zeta is elementary, $\zeta(-\ell, x) = -B_{\ell+1}(x)/(\ell + 1)$ with $B_m(x)$ the m -th Bernoulli polynomial [10, §25.11], so $\chi(\tau; X, Y)$ is a polynomial in τ of degree $n + 1$. Its role is that of a discrete antiderivative of the Eichler kernel: the Hurwitz shift $\zeta(-\ell, \tau) = \zeta(-\ell, \tau + 1) + \tau^\ell$ gives, term by term,

$$\chi(\tau - 1; X, Y) - \chi(\tau; X, Y) = (X - \tau Y)^n. \quad (36)$$

Both facts are used in the proof of Lemma 3.9.

For each τ , $\chi(\tau; \cdot, \cdot) \in V_{k-2} = \mathbb{C}[X, Y]_{k-2}$; the $\mathrm{SL}_2(\mathbb{Z})$ -slash $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$ acts on the polynomial variables alone, holding τ fixed:

$$\chi|_\gamma(\tau; X, Y) := \chi(\tau; aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \quad (37)$$

Thus $|_T$ sends $(X, Y) \mapsto (X + Y, Y)$ with τ untouched. Being linear and τ -independent, $|_\gamma$ commutes with $\int d\tau$, so slashing the polynomial E agrees with slashing χ under the integral; the two occurrences of $|_\gamma$ in Lemma 3.9 denote the same operation.

Lemma 3.9. *Let $f \in S_k^!$, $\tau_0 \in \mathbb{H}$, and let ω_f , $C_\gamma(f; \tau_0)$, $\chi(\tau; X, Y)$, and $E(f; \tau_0)$ be as defined in Eqs. (33), (34) and (35). Then $\hat{C}_\gamma := C_\gamma - (E|_\gamma - E)$ is a 1-cocycle for $\mathrm{SL}_2(\mathbb{Z})$ with $\hat{C}_S = r_f$ and $\hat{C}_T = 0$.*

Proof. First, we prove $\hat{C}_\gamma$ is a cocycle. Path-additivity and covariance of ω_f give the Eichler–Shimura cocycle relation $C_{\gamma\delta} = C_\gamma|_\delta + C_\delta$ [11, 6]; as $\gamma \mapsto E|_\gamma - E$ is a coboundary, $\hat{C}$ is a cocycle.

Having established it is a cocycle, we now show that $\hat{C}_S = r_f$. By Proposition 3.8 the period polynomial splits along the two segments of Definition 2.1, and we identify each contribution in turn. The S -segment carries the kernel τ^{s-1} , so assembling it over ℓ returns $(2\pi i)^{n+1} \int_{\gamma_{\tau_0}^S} \omega_f = C_S$ exactly.

For the T -segment, at $s = \ell + 1$ Definition 2.2 reads $\tilde{k}_T(\tau, \ell + 1) = \zeta(-\ell, \tau + 1) - (-1)^\ell \zeta(-(n - \ell), \tau + 1)$; assembling $\mathbf{K}_T$ and re-indexing $\ell \mapsto n - \ell$ in the second term (with n even) gives $\mathbf{K}_T = \chi - \chi|_S$, so the T -term of r_f is $E - E|_S$. Hence

$$r_f = C_S + (E - E|_S) = C_S - (E|_S - E) = \hat{C}_S. \quad (38)$$

Neither summand in (38) vanishes in general: the S -segment runs from $-1/\tau_0$ to τ_0 and degenerates only at the fixed point $\tau_0 = i$, where $C_S = 0$ and $E|_S - E = -r_f$.

Finally, we show that $\hat{C}_T = 0$. Applying the shift identity (36) twice shows that $A(\tau) := \chi(\tau; X + Y, Y) - \chi(\tau - 1; X, Y)$ satisfies $A(\tau - 1) = A(\tau)$. Crucially, χ is a polynomial in τ , as recorded below Eq. (35), so A is a 1-periodic polynomial and hence constant: $\partial_\tau A(\tau) = 0 \Leftrightarrow A(\tau) = A_0$. Thus,

$$\begin{aligned} \hat{C}_T &= C_T - (E|_T - E) \\ &= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)(X - \tau Y)^n d\tau - (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau)[\chi|_T(\tau; X, Y) - \chi(\tau; X, Y)] d\tau \\ &= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \left([\chi(\tau - 1; X, Y) - \chi(\tau; X, Y)] - [\chi(\tau; X + Y, Y) - \chi(\tau; X, Y)] \right) d\tau \\ &= (2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) [\chi(\tau - 1; X, Y) - \chi(\tau; X + Y, Y)] d\tau = -(2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f(\tau) A(\tau) d\tau \\ &= -(2\pi i)^{n+1} A_0 \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau = -(2\pi i)^{n+1} A_0 c_f(0) = 0, \end{aligned} \quad (39)$$

as all elements $f \in S_k^!$ must have $c_f(0) = 0$. $\square$

Definition 3.10. $W := \{w \in \mathbb{C}[X, Y]_{k-2} : w|_{(1+S)} = w|_{(1+U+U^2)} = 0\} \subset V_n$

Lemma 3.11. *For $f \in S_k^!$, and $W \subset V_n$ from Definition 3.10, the period polynomial $r_f(X, Y) \in W$.*

Proof. Let $\hat{C}_\gamma$ be the parabolic cocycle of Lemma 3.9. Thus, $r_f = \hat{C}_S$ and $\hat{C}_T = 0$. Now, note that $\hat{C}_\gamma(f; \tau_0) = 0$ when $\gamma = \pm I$. Since $n = k - 2$ is even, $\pm I$ act trivially on V_n , so the coboundary term $E|_{\pm I} - E$ vanishes and $\hat{C}_{\pm I} = C_{\pm I}$; and $C_{\pm I}$ is a closed-contour integral of a function which has no poles in $\mathbb{H}$. Finally, direct computation, starting from the identities $-I = U^3 = S^2$ and using the cocycle law $\hat{C}_{\gamma\delta} = \hat{C}_\gamma|_\delta + \hat{C}_\delta$ and clever use of $\hat{C}_T = 0$, shows $r_f|_{(1+S)} = r_f|_{(1+U+U^2)} = 0$:

$$\hat{C}_U = \hat{C}_{TS} = \hat{C}_T|_S + \hat{C}_S = \hat{C}_S = r_f, \quad (40)$$

$$0 = \hat{C}_{S^2} = \hat{C}_S|_S + \hat{C}_S = \hat{C}_S|_{(1+S)} = r_f|_{(1+S)}, \quad (41)$$

$$0 = \hat{C}_{U^3} = \hat{C}_U|_{UU} + \hat{C}_{UU} = \hat{C}_U|_{U^2} + (\hat{C}_U|_U + \hat{C}_U) = \hat{C}_U|_{(1+U+U^2)} = r_f|_{(1+U+U^2)}. \quad (42)$$

Thus, $r_f \in W \subset V_n$, and finishes the proof. $\square$

To define periods and quasi-periods, we recall the standard fact that $\dim(W) = 2 \dim(S_k) + 1$ (see, e.g. [6]), and define the inner-product:

$$\langle P, Q \rangle := \sum_a (-1)^a \binom{n}{a}^{-1} p_a q_{n-a} . \quad (43)$$

Now for the basis elements for k where $\dim(S_k) = 1$:

Definition 3.12. Let $\dim S_k = 1$, $n := k - 2$, $p_0(X) := X^n - 1$, $B_\ell := \frac{\ell!}{2\pi i} \oint \frac{t^{-\ell}}{e^t - 1} dt$, and $B_m^0(X) := \sum_{i \neq 1} \binom{m}{i} B_i X^{m-i}$. Motivated by [6, Thm. 1'], we define:

$$W_-^{(k)}(X) := \text{pp} \left(\frac{1}{n-1} (B_{n-1}^0(X) + X^n B_{n-1}^0(\frac{1}{X})) - \frac{1}{3} (B_3^0(X) + X^n B_3^0(\frac{1}{X})) \right), \quad (44)$$

$$W_+^{(k)}(X) := \text{pp} \left(\frac{1}{2} (B_2^0(X) - X^n B_2^0(\frac{1}{X})) + \frac{1}{n} (B_n^0(X) - X^n B_n^0(\frac{1}{X})) - c p_0(X) \right), \quad (45)$$

$$v_0^{(k)}(X) := p_0(X), \quad (46)$$

where pp acts on any polynomial with rational coefficients by multiplying by all coefficient-denominators and dividing by the gcd, and $\langle P, Q \rangle$ is as in Eq. (43). The bracketed polynomial defining $W_+^{(k)}$ has coefficients $a_i = -a_{n-i}$, so the single constant $c := a_n$ eliminates its X^n and constant terms at once; this is the p_0 -component, and removing it is what makes $\langle W_+^{(k)}, p_0 \rangle = 0$ below.

Lemma 3.13. For k with $\dim S_k = 1$, the space W of Lemma 3.11 has $\dim W = 3$, and $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$ is an orthogonal basis of W under $\langle \cdot, \cdot \rangle$ from Eq. (43).

Proof. First, let $W_\pm^{(k)}(X, Y) := Y^{k-2} W_\pm^{(k)}(X/Y)$, and let $v_0^{(k)}(X, Y) := Y^{k-2} v_0^{(k)}(X/Y)$.

Membership in W is the substantive point. Theorem 1' of [6] gives the odd and even period polynomials of the cusp form $E_{k-12}\Delta$ (where $E_0 := 1$) as modified-Bernoulli combinations and places them in W ; the $W_\pm^{(k)}$ of Definition 3.12 are specializations of those formulae, $W_-^{(k)}$ odd and $W_+^{(k)}$ even under $X \mapsto -X$, so both lie in W . So does $v_0^{(k)} = p_0 = X^n - Y^n$, directly: $p_0|_S = -p_0$ since n is even, and $p_0|_{(1+U+U^2)} = 0$.

By Eichler–Shimura $\dim W = 2 \dim S_k + 1 = 3$, so it is enough that the three be independent. Each satisfies $\langle P, P \rangle \neq 0$ [9], and mutually orthogonal vectors with $\langle P, P \rangle \neq 0$ are independent: pairing $\sum_i c_i v_i = 0$ against v_j leaves $c_j \langle v_j, v_j \rangle = 0$.

For orthogonality, note first that since n is even, $X \mapsto -X$ multiplies the coefficient of $X^{n-a} Y^a$ by $(-1)^a$, so an odd polynomial has $p_a = 0$ for even a and an even one has $q_b = 0$ for odd b . As Eq. (43) pairs p_a against q_{n-a} , and a odd forces $n - a$ odd, opposite parities pair to zero: the odd $W_-^{(k)}$ is orthogonal to the even $W_+^{(k)}$ and $v_0^{(k)}$. For the remaining pair, p_0 is supported on its two extreme coefficients, so $\langle P, p_0 \rangle$ is the coefficient of Y^n in P less that of X^n ; c annihilates both of these in $W_+^{(k)}$, whence $\langle W_+^{(k)}, v_0^{(k)} \rangle = 0$. Hence $\{W_+^{(k)}, W_-^{(k)}, v_0^{(k)}\}$ is an orthogonal basis, with $\langle v_0^{(k)}, v_0^{(k)} \rangle = -2$ at every such k . $\square$

Definition 3.14 (Periods, Quasi-Periods). Let k be such that $\dim(S_k) = 1$, $n := k - 2$, $W_\pm^{(k)}(X, Y) := Y^n W_\pm^{(k)}(X/Y)$ be as in Definition 3.12, and $r_f(X, Y)$ be as in Definition 3.7. Now, define

$$\omega^\pm(f) := \frac{\langle r_f, W_\pm^{(k)} \rangle}{\langle W_\pm^{(k)}, W_\pm^{(k)} \rangle}, \quad (47)$$

where $\langle \cdot, \cdot \rangle$ is as in Eq. (43). When f is a cusp form, $\omega^\pm(f)$ are its periods; when f is a weak cusp form, $\omega^\pm(f)$ are its quasi-periods, sometimes denoted $\eta^\pm(f)$.

Now, having defined our periods and quasi-periods, we use them together with our L-functions to extract periods and quasi-periods for $f \in S_k^!$ (when $\dim(S_k) = 1$). To this effect, we recall the notation of [5]. Decomposing the modular weight $k = 12\lambda + k'$ where $k' \in \{0, 4, 6, 8, 10, 14\}$, we see $\lambda = \dim(S_k)$ when $\lambda \geq 0$. For any λ , we may denote $f_{k,m} := 1/q^m + O(q^{\lambda+1}) \in M_k^!$. When $\dim(S_k) = 1$, we have $f_{k,m} = 1/q^m + O(q^2)$. Here, define $\Delta_k := f_{k,-1} = q + O(q^2) \in S_k$ and $\hat{\Delta}_k := f_{k,1} = 1/q + O(q^2) \in S_k^!$. Further, define

$$\text{pd}_k := \begin{pmatrix} \omega^+(\Delta_k) & \omega^-(\Delta_k) \\ \eta^+(\hat{\Delta}_k) & \eta^-(\hat{\Delta}_k) \end{pmatrix}, \quad \alpha_k := \frac{(-1)^{1+k/2} \det(\text{pd}_k)}{(2\pi i)^{k-1} (k-2)!} \quad (48)$$

k	$\omega^+(\Delta_k)$	$\omega^-(\Delta_k)/i$	$\eta^+(\hat{\Delta}_k)$	$\eta^-(\hat{\Delta}_k)/i$	α_k
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	1
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	13/30
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	2/3
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	17/5
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	19/21
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	23/15

Table 1: Periods $\omega^\pm(\Delta_k)$ and quasi-periods $\eta^\pm(\hat{\Delta}_k)$, and the rational number α_k . While $\alpha_{12} = 1$ matches Brown [2], α_k for $k \geq 16$ appear to be new, interesting, rational numbers, albeit with implicit dependence on the period-polynomial basis normalizations—in contrast to the normalization-free Petersson check above. We have not found a clear pattern for the α_k . See [9] for the scripts that extracted these figures (to 50-digit precision).

Brown–Hain [7, Prop. 5.6] show that $\det(\text{pd}_k)$ satisfies a Legendre-type relation, $\det(\text{pd}_k) \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$. Brown [2, §8.3] numerically finds $\alpha_k = 1$; we reproduce this result at $k = 12$ and give the analogous rational multiples for $k \in \{16, 18, 20, 22, 26\}$ in Table 1, below, via the normalization of Definition 3.12.

4 L-functions, period polynomials and poles at generic locations

For the reconstruction of period polynomials from special values of L^* , which Section 3 carried out for $f \in S_k^!$, we needed a region free of poles. As it happens, and as we show below, when the L-functions defined in this section are used to build period polynomials that are naturally elements of W , choosing the base point $\tau_0 = \rho + 1 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and taking the base contour of the reference homotopy class to lie on the arc $|\tau| = 1$ running from ρ to $\rho + 1$, along the lower boundary of the fundamental domain and its S -image, makes that pole-free region of codimension zero in the upper half-plane. Motivated by this, we derive here explicit expressions for the L-function along this reference contour, defined so as to allow similarly explicit representations in terms of the q -series coefficients of generic $f \in F_k$.

For $f \in F_k$ with poles in $\mathbb{H}$, the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

Proposition 4.1. *For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, the L-integral $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends only on the homotopy class of its two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, and not on the basepoint τ_0 . At fixed τ_0 each of L_S^*, L_T^* is separately path-independent.*

Proof. At fixed τ_0 the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so Cauchy’s theorem makes each of L_S^*, L_T^* constant on the homotopy class of its own segment. Under

a motion of the basepoint, $\partial_{\tau_0} L^*$ is evaluated at the moving endpoints alone; the computation of Proposition 2.6, which uses only $f(\tau_0 - 1) = f(\tau_0)$ and $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, cancels the boundary terms of L_S^* against those of L_T^* , the interior poles lying off the contours. Hence $\partial_{\tau_0} L^* = 0$ along any motion in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, while each summand may separately vary. $\square$

Definition 4.2. For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, let $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$ denote the homotopy classes of the two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Thus, with $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$ and $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$, by Proposition 4.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (49)$$

is independent of the chosen representative paths and independent of τ_0 (subject to natural constraints of τ_0 monodromies around singular points, which move between homotopy classes).

The independence of τ_0 is exact, and the mechanism is the T -shift of the kernel; we record it, since the treatment of a pole at an elliptic point rests on it.

Lemma 4.3. Let $f \in F_k$ and let $\tau_0, \tau'_0 \in \mathbb{H}$ be joined by a path σ meeting no pole of f , with the segments at τ'_0 obtained from those at τ_0 by appending σ and its S - and T^{-1} -images. Then $L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ for every $s \in \mathbb{C}$.

Proof. Write Δ for the change in each segment integral. On the S -side, $\Delta L_S^* = -\int_{\sigma} f(\tau) \tau^{s-1} d\tau + \int_{S\sigma} f(\tau) \tau^{s-1} d\tau$, and substituting $\tau \mapsto S\tau$ with $f|_k S = f$ gives $f(S\tau)(S\tau)^{s-1} d(S\tau) = (-1)^{s-1} f(\tau) \tau^{k-s-1} d\tau$, so the second term is $(-1)^{s-1} \int_{\sigma} f(\tau) \tau^{k-s-1} d\tau$. On the T -side, T -periodicity of f gives $\Delta L_T^* = -\int_{\sigma} f(\tau) [\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s)] d\tau$, and $\zeta(w, \tau + 1) = \zeta(w, \tau) - \tau^{-w}$ applied to both Hurwitz terms of Definition 2.2 yields

$$\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{k-s-1}. \quad (50)$$

Adding the two changes, the integrand is $-\tau^{s-1} + (-1)^{s-1} \tau^{k-s-1} + \tau^{s-1} - e^{i\pi(s-1)} \tau^{k-s-1}$, which vanishes identically since $(-1)^{s-1} = e^{i\pi(s-1)}$. The prefactor $e^{-\pi i s/2}$ multiplies through. $\square$

The hypothesis is only that σ avoids the poles: crossing one moves between homotopy classes, which is the content of Lemma 4.8 below. In particular L^* is *constant* in any parameter that slides τ_0 without crossing a pole, not merely convergent as that parameter tends to a limit.

We would like to speak of “the” L-integral, and homotopy classes split this into L-integrals for contours in different classes. We therefore fix a reference class, and we fix it on the boundary of the fundamental domain.

Definition 4.4 (Reference homotopy class for $f \in F_k$). Let γ^{arc} be the geodesic segment $\{e^{i\theta} : \frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}\}$, oriented from ρ to $\rho + 1$, and let $\tau_0 = \rho + 1$. The geodesic reference class is the class of $\hat{\gamma}_f^S = \hat{\gamma}_f^T = \gamma^{\text{arc}}$.

The choice is admissible by Definition 2.1, since $-1/(\rho + 1) = \rho$ and $(\rho + 1) - 1 = \rho$: the two segments share their endpoints and may be taken to be the same path.

Poles on γ^{arc} never occur singly. As S acts on the unit circle by $e^{i\theta} \mapsto e^{i(\pi-\theta)}$, preserving $\frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}$, while $f^{-1}(\infty)$ is Γ -stable and $f|_k S = f$, a pole of order P at $e^{i\phi}$ on the arc is accompanied by one of the same order at $e^{i(\pi-\phi)}$, again on the arc. These S -pairs have two degenerations: the pair collapses to a single point exactly when $\phi = \frac{\pi}{2}$, and it consists of the two endpoints exactly when $\phi \in \{\frac{\pi}{3}, \frac{2\pi}{3}\}$, since $S(\rho + 1) = -1/(\rho + 1) = \rho$. All three cases are treated the same way, by moving the poles off the contour rather than moving the contour.

We move the contour off the poles, never the poles off the contour. Displacing a pole is not available to us: $f^{-1}(\infty)$ is $\mathrm{SL}_2(\mathbb{Z})$ -stable, so a pole cannot be moved alone, and at an elliptic point of order n the stabiliser acts as rotation by $2\pi/n$ about a point the contour runs through, sending any displacement to the far side and pinching the contour. Displacing the base point costs nothing, by Lemma 4.3.

Definition 4.5. For $0 < \epsilon < \frac{1}{2}$ put $\tau_0(\epsilon) := (1 + \epsilon)e^{i\pi/3}$, so that $S\tau_0(\epsilon) = (1 + \epsilon)^{-1}e^{2i\pi/3}$ and $T^{-1}\tau_0(\epsilon) = \rho + \epsilon e^{i\pi/3}$. Let $\hat{\gamma}_\epsilon^S$ be the logarithmic spiral

$$\tau(\theta) = \exp\left(\log(1 + \epsilon) \frac{\pi/2 - \theta}{\pi/6} + i\theta\right), \quad \theta : \frac{2\pi}{3} \rightarrow \frac{\pi}{3}, \quad (51)$$

and let $\hat{\gamma}_\epsilon^T$ be a path from $T^{-1}\tau_0(\epsilon)$ to $S\tau_0(\epsilon)$ inside $|\tau - \rho| = O(\epsilon)$, followed by $\hat{\gamma}_\epsilon^S$. The ϵ -shifted geodesic class is the pair of their homotopy classes in $\mathbb{H} \setminus f^{-1}(\infty)$.

Three features of (51) are used below. Its $\log r$ is odd about $\theta = \frac{\pi}{2}$ and S acts by $(r, \theta) \mapsto (1/r, \pi - \theta)$, so $\hat{\gamma}_\epsilon^S$ is S -symmetric and Lemma 4.26 applies to the S^2 -path. It lies outside $|\tau| = 1$ for $\theta < \frac{\pi}{2}$ and inside for $\theta > \frac{\pi}{2}$, so an S -pair of poles on the arc is passed on opposite sides automatically, with no side chosen by hand. And it reduces to γ^{arc} as $\epsilon \rightarrow 0$, the two segments then sharing their endpoints as in Definition 4.4.

Since $S\tau_0(\epsilon)$ and $T^{-1}\tau_0(\epsilon)$ are distinct for $\epsilon \neq 0$, the connector in Definition 4.5 is not determined by its endpoints, and the classes available to it are indexed by an integer. We name that integer through the relator path, where it is what the period polynomial sees.

Definition 4.6. Let $W(f, \epsilon) \in \mathbb{Z}$ be the winding number about $\rho + 1$ of the closed path traced by $(ST)^3$ in the ϵ -shifted geodesic class, computed in the coordinate $w = (\tau - \rho - 1)/(\tau - \overline{\rho + 1})$.

By Lemma 4.8 the L-integral is an affine function of W , and by Lemma 4.32 below $W \equiv -1 \pmod{3}$, so $W = 0$ is not realised by any single contour. It is nonetheless the value at which the period polynomial lies in W , and L^* is linear, so we define the L-integral by evaluating that affine function there.

Definition 4.7. Let $f \in F_k$ have a pole on $\mathrm{SL}_2(\mathbb{Z}) \cdot \rho$ or at i . Then $L^*(f, s)$ is the value at $W = 0$ of the affine function $W \mapsto L^*(f, s; \hat{\gamma}_\epsilon^S, \hat{\gamma}_\epsilon^T)$, and likewise for the period polynomial of Definition 3.7.

Equivalently, $L^*(f, s)$ is the weighted average of L-integrals over two realisable classes, with weights w_j fixed by $\sum_j w_j = 1$ and $\sum_j w_j W_j = 0$: at i , where W is odd, the two classes $W = \pm 1$ enter with weight $\frac{1}{2}$ each, so L^* is the mean of the two indentations; at ρ the classes $W = -1$ and $W = 2$ enter with weights $\frac{2}{3}, \frac{1}{3}$. In both cases the correction to the non-winding class is $\frac{1}{n}$ of one residue,

$$L^*(f, s) = L^*(f, s; W = -1) - \frac{2\pi i}{n} e^{-\pi i s/2} \mathrm{Res}_{\tau=\tau_e}(f \tilde{k}_T), \quad (52)$$

the weight $\frac{1}{n}$ depending only on the order of the elliptic point, never on f or on the order of its pole. So (52) is linear in f , and is independent of ϵ by Lemma 4.3. Poles of f on γ^{arc} away from the elliptic points need no prescription at all: the spiral of (51) already separates each S -pair, and by Lemma 4.30 either separation puts $\hat{r}_f$ in W .

With this definition we may unambiguously write

$$L^*(f, s) := L^*(f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]), \quad (53)$$

where for f with a pole at an elliptic point the right-hand side is read through Definition 4.7: there the value is not an integral over one homotopy class but the weighted average of two, so (53) fixes a

number and not a contour. Having fixed this reference value, consider two paths $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ which may lie in any homotopy class, and let $X_S(p), X_T(p) \in \mathbb{Z}$ count the windings of $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ about each pole $\tau_p \in f^{-1}(\infty)$ relative to the reference class.

Lemma 4.8. *Let $f \in F_k$, $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$ and $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$. Then,*

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} [X_S(\tau_p)r_S(f, \tau_p) + X_T(\tau_p)r_T(f, \tau_p)]. \quad (54)$$

Proof. Two pairs whose winding data differ by one unit at a single pole p , on one segment, bound a loop encircling p once; the residue theorem adds $2\pi i$ times the residue of that segment's integrand, $f(\tau)\tau^{s-1}$ for γ^S or $f(\tau)\tilde{k}_T(\tau, s)$ for γ^T , and the prefactor $e^{-\pi i s/2}$ of Definition 2.3 multiplies through. Summing over crossings gives (54). $\square$

In particular two contours passing on opposite sides of a pole differ by its residue, and nothing else. For a pole on the arc away from the elliptic points this is the whole of the ambiguity.

Proposition 4.9. *Let $f \in F_k$ have poles on γ^{arc} , none of them at i , ρ or $\rho + 1$. Write $\gamma_{\pm}^{\text{arc}}(p)$ for γ^{arc} indented near p so as to pass outside, respectively inside, $|\tau| = 1$. Then $L^*(f, s)$ is defined on either contour, and replacing $\gamma_+^{\text{arc}}(p)$ by $\gamma_-^{\text{arc}}(p)$ at one pole changes it by*

$$-2\pi i e^{-\pi i s/2} [r_S(f, p) + r_T(f, p)]. \quad (55)$$

The spiral $\hat{\gamma}_\epsilon^S$ of (51) realises one such contour for every $\epsilon > 0$, passing outside at $\theta < \frac{\pi}{2}$ and inside at $\theta > \frac{\pi}{2}$, so each S -pair is separated without a side being chosen by hand.

Proof. Each indented contour keeps p at positive distance, so both segment integrals converge and L^* is defined on either; the spiral does the same for every $\epsilon > 0$, and the sides it takes are as stated because $\log r$ in (51) is positive for $\theta < \frac{\pi}{2}$ and negative for $\theta > \frac{\pi}{2}$.

For the difference, $\gamma_+^{\text{arc}}(p)$ and $\gamma_-^{\text{arc}}(p)$ agree off a neighbourhood of p and together bound a loop encircling p once. Along γ^{arc} the direction of travel is θ decreasing, so it is $-ie^{i\theta}$ up to a positive factor, while the outward radial direction is $e^{i\theta} = i \cdot (-ie^{i\theta})$: outward lies to the left of travel. The sliver between the two indentations therefore lies to the right of $\gamma_+^{\text{arc}}(p)$ traversed forward, and the loop encircles p negatively. This is the winding datum $X_S(p) = X_T(p) = -1$ of Lemma 4.8, which gives (55). $\square$

The exclusion of i , ρ and $\rho + 1$ is not a technicality: there the stabiliser orbit meets γ^{arc} on both sides at once, no single-sided indentation exists, and the winding of Definition 4.6 cannot be set to zero by any one contour.

This provides the basics for the finite L-functions of $f \in F_k$ in this class. We wish, however, to present them in terms of their q -series coefficients, as was done for $f \in M_k^!$ in Section 3. The obstruction to doing so directly is not a technical one. For $f \in M_k^!$ the series $\sum_n c_f(n)q^n$ converges on all of $\mathbb{H}$, as f is holomorphic there, and term-by-term evaluation is legitimate throughout. A pole at finite height ends this. The expansion about the cusp then represents f only above the highest pole, on $\text{Im } \tau > y_f := \sup\{\text{Im } \tau_p\}$, with coefficients growing as $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$ for a pole of order m [3]. The arc has height varying between $\frac{\sqrt{3}}{2}$ and 1, so the operative threshold is its lowest point $\frac{\sqrt{3}}{2}$: as soon as f has a pole at or above that height the q -series diverges somewhere on the contour, and there is then no series to integrate term by term, rather than a slowly convergent one to be handled with care. To do this, we must introduce the polylogarithmic projections of Ref. [3].

4.1 Explicit formulae for $L^*(f, s)$ on the arc

We lay out the starting definitions and facts.

Definition 4.10. $F(\infty)$ is the space of meromorphic functions $g : \mathbb{H} \rightarrow \mathbb{C}$ with $g(\tau + 1) = g(\tau)$, finitely many poles in each compact subset of the strip $|\operatorname{Re} \tau| \leq \frac{1}{2}$, and bounded pole order at $q = 0$.

Definition 4.11. Let $f \in F_k$ have a pole of order m_p at $\tau_p \in \mathbb{H}$. On a punctured ε -ball about τ_p small enough to exclude the other poles of f , we have

$$f(\tau) = \sum_{m=1}^{m_p} \frac{r_{f,\tau_p}^*(m)}{(\tau - \tau_p)^m} + g(\tau), \quad r_{f,\tau_p}^*(m) = \frac{1}{2\pi i} \oint_{|\tau - \tau_p|=\varepsilon} f(\tau) (\tau - \tau_p)^{m-1} d\tau, \quad (56)$$

with g holomorphic at τ_p . We call $r_{f,\tau_p}^*(1), \dots, r_{f,\tau_p}^*(m_p)$ the pole weights of f at τ_p ; the first is the residue, and the last is nonzero, which is what fixes the order. The polylog projection of f at τ_p is

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \operatorname{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}), \quad \operatorname{Li}_s(x) := \sum_{n \geq 1} \frac{x^n}{n^s}. \quad (57)$$

The composite projector is the one place where the contour enters. Its threshold is set so that $(1 - \hat{P})f$ admits a q -series everywhere it is used, which is a statement about a half-plane: a q -series converges exactly on $\operatorname{Im} \tau > h$ with h the height of the highest surviving pole. The operative height is the lowest point of γ^{arc} , namely $\frac{\sqrt{3}}{2}$, since the mode sums below are controlled by $e^{-2\pi n \operatorname{Im} \tau}$ and the arc descends to that height at its endpoints.

Definition 4.12. For $f \in F_k$ with T -inequivalent finite-order poles $\{\tau_p\}$ of orders $\{m_p\}$, set

$$\hat{P}^{\text{geo}} := \sum_{\operatorname{Im} \tau_p \geq \frac{\sqrt{3}}{2}} \hat{P}_{\tau_p}, \quad (58)$$

with $\hat{P}_{\tau_p}$ as in Definition 4.11, and write $f = \hat{P}^{\text{geo}} f + (I - \hat{P}^{\text{geo}})f$.

Lemma 4.13. The normalized block $\frac{(-2\pi i)^m}{(m-1)!} \operatorname{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has principal part exactly $(\tau - \tau_p)^{-m}$, so $\hat{P}_{\tau_p} f$ matches the principal part of f and $(I - \hat{P}_{\tau_p})f$ is holomorphic at τ_p .

Proof. By the small- t expansion $\operatorname{Li}_{1-m}(e^{-t}) = (m-1)!t^{-m} + O(t^0)$ with $t = -2\pi i(\tau - \tau_p)$, the block has a pure pole $(\tau - \tau_p)^{-m} + O((\tau - \tau_p)^0)$ at τ_p , with no lower-order poles. Summing against the Laurent coefficients $r_{f,\tau_p}^*(m)$ reproduces the principal part of f term by term, leaving $(I - \hat{P}_{\tau_p})f$ regular at τ_p . $\square$

Definition 4.14. For $f \in F_k$ and $\hat{P}^{\text{geo}}$ as in Definition 4.12, write

$$(I - \hat{P}^{\text{geo}})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (59)$$

Lemma 4.15. $F_k \subset F(\infty)$ and $(I - \hat{P}^{\text{geo}})f \in F(\infty)$. Every pole of $(I - \hat{P}^{\text{geo}})f$ has $\operatorname{Im} \tau < \frac{\sqrt{3}}{2}$, with a maximal height $y_b < \frac{\sqrt{3}}{2}$ (set $y_b = 0$ if $(I - \hat{P}^{\text{geo}})f$ is holomorphic).

Proof. A weakly holomorphic form is T -periodic, meromorphic, and $O(e^{C \operatorname{Im} \tau})$ at the cusp, so $F_k \subset F(\infty)$; the projected polylogs $\operatorname{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ share these properties, so $(I - \hat{P}^{\text{geo}})f \in F(\infty)$ (and leaves F_k , being non-modular). By Lemma 4.13, $\hat{P}^{\text{geo}}$ cancels the principal part of every pole with $\operatorname{Im} \tau \geq \frac{\sqrt{3}}{2}$, so the poles of $(I - \hat{P}^{\text{geo}})f$ lie strictly below $\frac{\sqrt{3}}{2}$; finite in each compact set and accumulating only at $\operatorname{Im} \tau = 0$, they have a maximal height $y_b < \frac{\sqrt{3}}{2}$. $\square$

The Fourier coefficients of a T -periodic meromorphic function are controlled by its highest pole: $|c_{Rf}(n)| \leq e^{2\pi n(y_b + \varepsilon)}$ for every $\varepsilon > 0$ and all large n [3]. Since $y_b < \frac{\sqrt{3}}{2}$, the series $\sum_n c_{Rf}(n)q^n$ converges geometrically down to $\text{Im } \tau = \frac{\sqrt{3}}{2}$, at base $e^{-2\pi(\frac{\sqrt{3}}{2} - y_b - \varepsilon)}$, and so on γ^{arc} itself. This is the point of the projection: convergence merely for large $\text{Im } \tau$ would be automatic and of no use, as the contour descends to height $\frac{\sqrt{3}}{2}$.

One further integral of f along the contour will be needed, both for the residue of L^* at $s = 0$ and, later, for period polynomials.

Definition 4.16. For $f \in F_k$, set $\Phi(f) := \int_{\hat{\gamma}_f^T} f(\tau) d\tau$, the integral of f along the T -segment of the contour defining $L^*(f, s)$.

Φ is not an independent choice: it is what the parabolicity condition $\hat{C}_T = 0$ measures, and $\hat{C}_T$ is assembled from the T -transport, so Φ follows the contour. Deforming the segments replaces γ^{arc} in the integral by whatever $\hat{\gamma}_f^T$ then is, with no further stipulation.

Since $\hat{\gamma}_f^T$ runs from $T^{-1}\tau_0$ to τ_0 , and $e^{2\pi i n \tau}$ takes equal values at those two endpoints, the modes $n \neq 0$ of the q -series of f cancel between them. Hence $\Phi(f) = c_f(0)$ whenever that series converges along the segment, which is to say whenever no pole of f lies above $\hat{\gamma}_f^T$; a pole below is harmless. Otherwise $\Phi(f)$ is $c_f(0)$ less $2\pi i$ times the residues of the poles in the strip between the segment and the cusp. In particular $\Phi(E_k) = 1$, and $\Phi(f) = 0$ for $f \in S_k$.

The relations satisfied by the pole weights at an elliptic point are contour-free, so they hold here verbatim; we record them now, as both the block integral below and the period polynomials constructed later use them.

Lemma 4.17. Let τ_e be i or ρ , of order $n = 2$ or $n = 3$, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the generator S or ST of its stabiliser in $\text{SL}_2(\mathbb{Z})$, and set $\zeta := c\tau_e + d$, so that $c = 1$ and $\zeta = e^{i\pi/n}$. Let $f \in F_k$ have a pole of order P at τ_e with pole weights $r_{f, \tau_e}^*(m)$. Modularity is equivalent to one relation for each principal-part order μ ,

$$\sum_{m=\mu}^P r_{f, \tau_e}^*(m) \binom{m}{\mu} \zeta^m = \zeta^k \sum_{m=\mu}^P r_{f, \tau_e}^*(m) \binom{k}{m-\mu} \zeta^{-m}, \quad \forall \mu = 1, \dots, P, \quad (60)$$

and the leading relation is $(\zeta^{2P} - \zeta^k) r_{f, \tau_e}^*(P) = 0$, so that $2P \equiv k \pmod{2n}$.

Proof. Write $\tau = \tau_e + \varepsilon$. Since $\det \gamma = 1$ and $\gamma\tau_e = \tau_e$,

$$\gamma\tau - \tau_e = \frac{\tau - \tau_e}{(c\tau + d)(c\tau_e + d)} = \frac{\varepsilon}{\zeta(\zeta + \varepsilon)}, \quad (c\tau + d)^k = \zeta^k (1 + \varepsilon/\zeta)^k, \quad (61)$$

using $c = 1$. Hence $(\gamma\tau - \tau_e)^{-m} = \zeta^{2m} \varepsilon^{-m} (1 + \varepsilon/\zeta)^m$, and matching the coefficient of $\varepsilon^{-\mu}$ in $f(\gamma\tau) = (c\tau + d)^k f(\tau)$ through $(1 + \varepsilon/\zeta)^a = \sum_l \binom{a}{l} \zeta^{-l} \varepsilon^l$ gives $\sum_{m \geq \mu} r^*(m) \zeta^{2m} \binom{m}{\mu} \zeta^{-(m-\mu)} = \zeta^k \sum_{m \geq \mu} r^*(m) \binom{k}{m-\mu} \zeta^{-(m-\mu)}$; dividing by ζ^μ yields (60). At $\mu = P$ only $m = P$ survives on each side. $\square$

At $n = 2$ the order condition forces P odd when $k \equiv 2 \pmod{4}$ and P even when $4 \mid k$; at $n = 3$ it reads $2P \equiv k \pmod{6}$.

Each pole with $\text{Im } \tau_p \geq \frac{\sqrt{3}}{2}$ contributes one projected polylog to $L^*(f, s)$, whose L-integral we now record.

Definition 4.18. For a pole of order $N + 1$ projected at z , let $\lambda_{N,z}$ denote the corresponding block of (57), and set $I_N^{\text{geo}}(s, z) := L^*(\lambda_{N,z}, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T])$, evaluated in the geodesic reference class of Definition 4.4.

The blocks need not be integrated on the arc. Both kernels τ^{s-1} and $\tilde{k}_T(\tau, s)$ are holomorphic on all of $\mathbb{H}$, so the only obstruction to deforming is the block's own poles at $z + \mathbb{Z}$, and one may evaluate on the horizontal chord from ρ to $\rho + 1$ instead. Let Λ be the open lens between the two, bounded above by γ^{arc} and below by the chord, so that $\Lambda \subset \{|\tau| < 1, \text{Im } \tau > \frac{\sqrt{3}}{2}, |\text{Re } \tau| < \frac{1}{2}\}$; at most one member of $z + \mathbb{Z}$ lies in it. If z itself lies on γ^{arc} the contour is indented past it as in Proposition 4.9. In either situation set $\varepsilon_z := 1$ when z lies between the contour and the chord, that is when $z \in \Lambda$, or when $z \in \gamma^{\text{arc}}$ and the indentation is the outward one $\gamma_+^{\text{arc}}(z)$; and $\varepsilon_z := 0$ otherwise. The deformation to the chord costs one residue exactly when $\varepsilon_z = 1$.

Lemma 4.19. *Let $\text{Im } z > \frac{\sqrt{3}}{2}$, set $\tau_0 = \rho + 1$, let $\gamma^*(s, x) := \sum_{\ell \geq 0} \frac{(-x)^\ell}{\ell! (\ell + s)}$, entire in x , read empty products below as 1, and set $(2\pi n)^w := e^{w(\log 2\pi |n| + i\pi)}$ for $n < 0$. With, for $n \in \mathbb{Z} \setminus \{0\}$,*

$$G_{s,k}^{\text{geo}}(n) := e^{-\pi i s/2} \left[\tau_0^s \gamma^*(s, -2\pi i n \tau_0) - \rho^s \gamma^*(s, -2\pi i n \rho) \right] + \frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}}, \quad (62)$$

$$G_{s,k}^{\text{geo}}(0) := e^{-\pi i s/2} \frac{\tau_0^s - \rho^s}{s} - \frac{(\tau_0/i)^s}{s} - i^k \frac{(\tau_0/i)^{k-s}}{k-s}, \quad (63)$$

$$X_N(s, z) := \frac{e^{-\pi i s/2}}{(2\pi i)^N} \left[(-1)^N z^{s-1-N} \prod_{j=0}^{N-1} (s-1-j) + \zeta(1-s+N, z+1) \prod_{j=0}^{N-1} (1-s+j) - e^{i\pi(s-1)} \zeta(1-k+s+N, z+1) \prod_{j=0}^{N-1} (1-k+s+j) \right], \quad (64)$$

we have

$$I_N^{\text{geo}}(s, z) = -\delta_{N,0} G_{s,k}^{\text{geo}}(0) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} G_{s,k}^{\text{geo}}(-n) + \varepsilon_z X_N(s, z), \quad (65)$$

the sum converging at rate $e^{-2\pi(\text{Im } z - \frac{\sqrt{3}}{2})}$.

Proof. Both kernels are holomorphic on $\mathbb{H}$, and the only poles of $\lambda_{N,z}$ are at $z + \mathbb{Z}$, so Cauchy's theorem moves both segment integrals to the chord at the cost of whatever residues lie between the two paths; by the paragraph above that is $\{z\}$ when $\varepsilon_z = 1$ and nothing otherwise. The chord is not a convenience: along γ^{arc} itself $\text{Im } \tau$ ranges over $[\frac{\sqrt{3}}{2}, 1]$, so for $\frac{\sqrt{3}}{2} < \text{Im } z < 1$ the comparison below would change sign partway along, whereas on the chord $\text{Im } \tau \equiv \frac{\sqrt{3}}{2}$ and it is a single global inequality.

On the chord $|e^{2\pi i(\tau-z)}| > 1$, so the defining series of Li_{-N} diverges and inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w) - \delta_{N,0}$ rewrites $\lambda_{N,z}$ as $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} q^{-n} - \delta_{N,0}$, uniformly convergent there since $\text{Im } z > \frac{\sqrt{3}}{2}$. Integrating term by term, $G_{s,k}^{\text{geo}}(-n) = L^*(q^{-n}, s)$ in this class: the last two terms of (62) are L_T^* by Lemma 3.3 at τ_0 , and the first is $L_S^* = e^{-\pi i s/2} \int_{\rho}^{\tau_0} e^{-2\pi i n \tau} \tau^{s-1} d\tau$, which on expanding $e^{-2\pi i n \tau}$ and integrating $\tau^{\ell+s-1}$ term by term gives $e^{-\pi i s/2} \sum_{\ell \geq 0} \frac{(-2\pi i n)^\ell}{\ell! (\ell + s)} [\tau_0^{\ell+s} - \rho^{\ell+s}]$, the first bracket of (62). Likewise $G_{s,k}^{\text{geo}}(0) = L^*(1, s)$. The rate follows from $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$ against $|G_{s,k}^{\text{geo}}(-n)| \sim e^{\pi \sqrt{3} n}$.

For the crossing term put $u = \tau - z$ and $v = 2\pi i u + t$. Then $\sum_{N \geq 0} \text{Li}_{-N}(e^{2\pi i u}) \frac{t^N}{N!} = \frac{e^v}{1-e^v} = -\sum_{m \geq 0} \frac{B_m(-v)^m}{m! v}$, of which only the $m = 0$ term $-1/v$ carries negative powers of u ; expanding

$-1/(2\pi i u + t) = -\sum_{r \geq 0} (-1)^r t^r (2\pi i u)^{-r-1}$ and reading off $t^N/N!$ gives

$$\lambda_{N,z}(\tau) = \frac{(-1)^{N+1} N!}{(2\pi i(\tau - z))^{N+1}} + O(1), \quad (66)$$

a lone singular term, so $\text{Res}_{\tau=z}(\lambda_{N,z} \kappa) = (-1)^{N+1} \kappa^{(N)}(z)/(2\pi i)^{N+1}$ for any κ holomorphic at z . Both paths run $\rho \rightarrow \rho + 1$ with the arc above, so together they bound Λ clockwise and the deformation contributes $-2\pi i$ times that residue for each kernel. With $\partial_z^N z^{s-1} = z^{s-1-N} \prod_{j=0}^{N-1} (s-1-j)$ and $\partial_z^N \zeta(w, z+1) = (-1)^N \zeta(w+N, z+1) \prod_{j=0}^{N-1} (w+j)$ applied to both terms of k_T , this is $X_N(s, z)$ of (64). $\square$

Corollary 3.5 is deliberately not used here: it folds the S -segment about i using modularity, and q^n is only T -periodic, so its S -integral must be taken directly. Nor does that segment drop out, as it would for a base point at which the two segments coincide: on the arc it contributes the first bracket of (62), and the threshold governing the convergence rate stands at $\frac{\sqrt{3}}{2}$, in step with Definition 4.12.

The S -term is written with γ^* because that form is branch-free: γ^* is single-valued, so the only multivalued factors are the explicit τ_0^s and ρ^s , and those are unambiguous since $\arg \tau \in [\frac{\pi}{3}, \frac{2\pi}{3}]$ along the chord. It may equally be written with upper incomplete gammas, provided the sheet relation is carried. Under $u = 2\pi i n \tau$ the chord's image runs from $\arg u = \frac{7\pi}{6}$ at ρ to $\frac{5\pi}{6}$ at τ_0 , so the ρ endpoint sits one turn off principal; since $\gamma(s, x) = x^s \gamma^*(s, x)$ with γ^* single-valued, a shift of $\arg x$ by 2π gives

$$\Gamma^{\text{cont}}(s, x) = e^{2\pi i s} \Gamma(s, x) + (1 - e^{2\pi i s}) \Gamma(s), \quad (67)$$

and hence

$$S_n(s) = (2\pi i n)^{-s} \left[e^{2\pi i s} \Gamma(s, 2\pi i n \rho) + (1 - e^{2\pi i s}) \Gamma(s) - \Gamma(s, 2\pi i n \tau_0) \right]. \quad (68)$$

Both the sheet factor and the additive constant are trivial at integer s , which is all that Definition 3.7 requires. The T -term needs neither, since by Lemma 3.3 only τ_0 appears there, at $\arg(2\pi i n \tau_0) = \frac{5\pi}{6}$, the ρ endpoint having been absorbed by the periodicity of $\tilde{k}_T$. It does need the branch of $(2\pi n)^w$ at $n < 0$ fixed above, which is the one making $\arg(2\pi n) + \arg(\tau_0/i) = \frac{5\pi}{6}$; the opposite choice differs by $e^{2\pi i s}$, and so is invisible at integer s and wrong elsewhere. In practice the γ^* series loses about $2.7n$ digits to cancellation, so (68) is preferable if large n is needed.

On the threshold line itself the roles of the two series are exchanged. A pole at $\text{Im } z = \frac{\sqrt{3}}{2}$ lies below the arc, which at abscissa $\text{Re } z$ has height $\sqrt{1 - (\text{Re } z)^2} > \frac{\sqrt{3}}{2}$, so it is the defining series of Li_{-N} that applies rather than its inversion. That series converges absolutely for $\text{Im } z < \frac{\sqrt{3}}{2}$ and is exactly marginal at $\frac{\sqrt{3}}{2}$, since the arc attains that height at its two endpoints. Write $(a)_{(\ell)} := a(a-1)\cdots(a-\ell+1)$ for the falling factorial, with $(a)_{(0)} := 1$.

Lemma 4.20. *Let $\text{Im } z = \frac{\sqrt{3}}{2}$, and either $z \not\equiv \rho \pmod{\mathbb{Z}}$ or $c_N(s, k) = 0$, and set*

$$c_\ell(s, k) := (s-1)_{(\ell)} \left[-i i^\ell e^{-\pi i s/2} (\tau_0^{s-1-\ell} - \rho^{s-1-\ell}) + (-i \tau_0)^{s-1-\ell} \right] + i^k (k-s-1)_{(\ell)} (-i \tau_0)^{k-s-1-\ell}. \quad (69)$$

Then for every $L \geq N$,

$$I_N^{\text{geo}}(s, z) = \sum_{\ell=0}^L \frac{c_\ell(s, k)}{(2\pi)^{\ell+1}} \text{Li}_{\ell+1-N}(e^{2\pi i(\tau_0-z)}) + R_{N,L}^{\text{geo}}(s, z), \quad (70)$$

where

$$R_{N,L}^{\text{geo}}(s, z) := \sum_{n \geq 1} n^N e^{-2\pi i n z} \left[G_{s,k}^{\text{geo}}(n) - e^{2\pi i n \rho} \sum_{\ell=0}^L \frac{c_\ell(s, k)}{(2\pi n)^{\ell+1}} \right] \quad (71)$$

converges absolutely, its terms being $O(n^{N-L-2})$.

Proof. For $\text{Im } z < \frac{\sqrt{3}}{2}$ the series $\lambda_{N,z} = \sum_{n \geq 1} n^N e^{-2\pi i n z} q^n$ converges uniformly on γ^{arc} , both kernels are bounded there, and term-by-term integration gives $I_N^{\text{geo}}(s, z) = \sum_{n \geq 1} n^N e^{-2\pi i n z} G_{s,k}^{\text{geo}}(n)$, absolutely convergent.

Repeated integration by parts of the S -integral gives $\int_\rho^{\tau_0} e^{2\pi i n \tau} \tau^{s-1} d\tau \sim \sum_{\ell \geq 0} \frac{(-1)^\ell (s-1)_\ell}{(2\pi i n)^{\ell+1}} [e^{2\pi i n \tau} \tau^{s-1-\ell}]_\rho^{\tau_0}$, while $\Gamma(a, X) \sim X^{a-1} e^{-X} \sum_{\ell \geq 0} (a-1)_\ell X^{-\ell}$ at $X = -2\pi i n \tau_0 = 2\pi n(-i\tau_0)$ handles the two incomplete gammas. Since $e^{2\pi i n \tau_0} = e^{2\pi i n \rho}$ by T -periodicity, both endpoints carry the same exponential and, using $(-1)^\ell / i^{\ell+1} = -i i^\ell$, $G_{s,k}^{\text{geo}}(n) = e^{2\pi i n \rho} [\sum_{\ell=0}^L c_\ell(s, k) (2\pi n)^{-\ell-1} + O(n^{-L-2})]$ with c_ℓ as in (69).

Splitting the sum along this expansion, the peeled part is $\sum_{\ell=0}^L \frac{c_\ell}{(2\pi)^{\ell+1}} \sum_{n \geq 1} n^{N-\ell-1} e^{2\pi i n(\rho-z)}$, and $e^{2\pi i(\rho-z)} = e^{2\pi i(\tau_0-z)}$, giving the polylogarithms of (70); the remainder is (71), whose terms are $O(n^{N-L-2})$ because $|e^{-2\pi i n z} e^{2\pi i n \rho}| = 1$ on the threshold line. Both sides of (70) are holomorphic in z wherever z avoids $\gamma^{\text{arc}} + \mathbb{Z}$ and $e^{2\pi i(\tau_0-z)} \neq 1$ — the left because the contour integral of Definition 4.18 then has no singular integrand, the right because $R_{N,L}^{\text{geo}}$ converges locally uniformly — so the identity, proved on $\text{Im } z < \frac{\sqrt{3}}{2}$, persists onto the threshold line away from $\rho + \mathbb{Z}$.

At $z \equiv \rho$ the argument $e^{2\pi i(\tau_0-z)}$ equals 1 and the $\ell = N$ term carries $\text{Li}_1(1)$, so the statement needs $c_N = 0$. That coefficient is exactly the endpoint mismatch: by the computation above, c_0 is a non-zero multiple of

$$[\rho^{s-1} + \tilde{k}_T(\rho, s)] - [\tau_0^{s-1} + \tilde{k}_T(\tau_0, s)], \quad (72)$$

which by the shift identity (50) reads $\rho^{s-1} = e^{i\pi(s-1)} \tau_0^{k-s-1}$ and so, with $\rho = e^{2\pi i/3}$ and $\tau_0 = e^{i\pi/3}$, vanishes for every s precisely when $k \equiv 2 \pmod{6}$. This is not a side condition to be hoped for. A pole at ρ has $P \equiv 1 \pmod{n}$ by Lemma 4.17, and at $P = 1$ — the only order for which the block itself, rather than the winding prescription of Definition 4.7, is what one needs — that forces $k \equiv 2 \pmod{6}$ and so $c_0 = 0$. The divergence one might have expected at ρ is absent exactly when the theory calls on it. $\square$

The blocks assemble into the projection at a pole by linearity alone.

Lemma 4.21. *Let $f \in F_k$ have a pole of order m_p at τ_p with $\text{Im } \tau_p > \frac{\sqrt{3}}{2}$, and pole weights $r_{f, \tau_p}^*(m)$ as in Definition 4.11. Then*

$$L^*(\hat{P}_{\tau_p} f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^{m_p} r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}^{\text{geo}}(s, \tau_p). \quad (73)$$

Proof. By (57) the projection $\hat{P}_{\tau_p} f$ is the finite combination $\sum_m r_{f, \tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \lambda_{m-1, \tau_p}$ of the blocks of Definition 4.18. Both segment integrals of Definition 2.3 are linear in the integrand and the combination is finite, so L^* distributes over it. $\square$

What survives every such projection carries a q -series on the whole contour, and may be integrated mode by mode.

Lemma 4.22. *Let $f \in F_k$, write $Rf := (I - \hat{P}^{\text{geo}})f$, and let $c_{Rf}(n)$ be its Fourier coefficients as in Definition 4.14. Then*

$$L^*(Rf, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = c_{Rf}(0) G_{s,k}^{\text{geo}}(0) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}^{\text{geo}}(n), \quad (74)$$

the sum converging absolutely.

Proof. By Lemma 4.15 every pole of Rf has height at most $y_b < \frac{\sqrt{3}}{2}$, so for any $\varepsilon > 0$ with $y_b + \varepsilon < \frac{\sqrt{3}}{2}$ one has $|c_{Rf}(n)| \leq e^{2\pi n(y_b + \varepsilon)}$ for all large n , while only finitely many $n < 0$ occur, Rf being meromorphic at the cusp. On γ^{arc} one has $\text{Im } \tau \geq \frac{\sqrt{3}}{2}$, so those terms are bounded by $e^{-2\pi n(\frac{\sqrt{3}}{2} - y_b - \varepsilon)}$ and $\sum_n c_{Rf}(n) q^n$ converges uniformly on the contour. Both kernels are continuous, hence bounded, on that compact set, so each segment integral may be taken term by term. The mode q^n with $n \neq 0$ contributes $L^*(q^n, s) = G_{s,k}^{\text{geo}}(n)$, by the computation in the proof of Lemma 4.19 read at general n , and $q^0 = 1$ contributes $G_{s,k}^{\text{geo}}(0)$ of (63). $\square$

Lemma 4.19 assumes $\text{Im } z > \frac{\sqrt{3}}{2}$ strictly, so it says nothing on the threshold line itself, where the chord meets the pole and neither the defining series of Li_{-N} nor its inversion converges. That line is treated separately below, away from ρ and at it.

4.2 Relations to the L-functions of [3]

We pause to connect the L-functions of this section to the definition in Ref. [3].

Remark 4.23. The $1/s$ residue is carried entirely by the T -segment. L_S^* is a finite integral of $f \tau^{s-1}$ and is entire in s , while $\tilde{k}_T$ has a simple pole at $s = 0$ whose coefficient is the integral of f along that segment, the constant Fourier mode of f in the strip containing it. In the geodesic class that integral is $\Phi(f)$ of Definition 4.16, so $\text{Res}_{s=0} L^*(f, s) = -\Phi(f)$, whereas Ref. [3] anchors its contour at the cusp, above every pole, and carries $-c_f(0)$. The two differ by the residues lying in the strip between the contours, which is Lemma 4.8. The quantity that must be subtracted to reach W is therefore the same object as the departure from the residue convention of [3], and not a second normalisation imposed on top of it.

Remark 4.24. The mechanism is already visible in (63), before any assembly. Its S -part $e^{-\pi i s/2}(\tau_0^s - \rho^s)/s$ is finite at $s = 0$, since $\tau_0^s - \rho^s$ vanishes like $s \log(\tau_0/\rho)$, while $-(\tau_0/i)^s/s$ carries the pole. The two classes sit at opposite ends of one deformation: the contour of [3] lies above every pole, making the residue bookkeeping trivial and the pole-free region empty, while γ^{arc} lies below every pole orbit not on it, making the pole-free region of codimension zero and the residue bookkeeping maximal.

4.3 Period polynomials in the geodesic class

Definition 4.25. *For $f \in F_k$, let r_f be given by Definition 3.7 with $L^*(f, s)$ evaluated in the geodesic reference class of Definition 4.4, and set*

$$\hat{r}_f := r_f - \Phi(f) r_{E_k}. \quad (75)$$

$\text{SL}_2(\mathbb{Z})$ is generated by S and T subject to $S^2 = (ST)^3 = -I$, so two closed paths control the relations. Their degeneration on the arc is a statement about paths alone, independent of f ; we record it separately from its consequence, since the case of a pole *on* the contour uses the geometry but not the consequence.

Lemma 4.26. *Let τ_0 lie on γ^{arc} and let $\hat{\gamma}^S = \hat{\gamma}^T$. Then the closed path traced by S^2 retraces itself, and if moreover $\tau_0 = \rho + 1$ the closed path traced by $(ST)^3$ retraces itself. Both statements are independent of f .*

Proof. S acts on the unit circle by $e^{i\theta} \mapsto e^{i(\pi-\theta)}$, preserving $\frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}$, reversing orientation and fixing i ; so $S\hat{\gamma}^S$ is $\hat{\gamma}^S$ reversed and the S^2 -path retraces, for any base point on the arc. For $(ST)^3$, the fixed points of $U = TS$ are the roots of $\tau^2 - \tau + 1$, namely $e^{\pm i\pi/3}$, so $U\tau_0 = \tau_0$ exactly at $\tau_0 = \rho + 1$ and the U -path is closed. With $\hat{\gamma}^S = \hat{\gamma}^T$ it is $S^{-1}\hat{\gamma}^T$ followed by $\hat{\gamma}^S$, the arc traversed and then retraced; its U - and U^2 -transports retrace $U\gamma^{\text{arc}}$ and $U^2\gamma^{\text{arc}}$ through the same fixed point. $\square$

Lemma 4.27. *In the geodesic reference class, $\hat{C}_{\pm I} = 0$ for every $f \in F_k$ with no pole on $\text{SL}_2(\mathbb{Z})\gamma^{\text{arc}}$.*

Proof. By Lemma 4.26 both relator paths retrace, so each bounds nothing; the winding number about every pole of f vanishes and Cauchy's theorem gives $\hat{C}_{\pm I} = C_{\pm I} = 0$. $\square$

Definition 4.25 covers those f whose poles avoid the contour, and since γ^{arc} is a curve that is the generic situation: the excluded locus is of codimension one in $\mathbb{H}$, so $\hat{r}_f$ is defined outright on a set of codimension zero. The remaining cases are reached by continuity from it. Three cases exhaust the possibilities:

1. f has no pole on $\text{SL}_2(\mathbb{Z})\cdot\gamma^{\text{arc}}$;
2. f has a pole on γ^{arc} , away from i and ρ ;
3. f has a pole at i or at ρ .

Lemma 4.28. *Let $f \in F_k$ have no pole on $\text{SL}_2(\mathbb{Z})\cdot\gamma^{\text{arc}}$. Then $\hat{r}_f \in W$, with no condition on the location or the order of the poles of f .*

Proof. Throughout, $\hat{C}$ is the cocycle of Lemma 3.9, so that $\hat{C}_S = r_f$ and $\hat{C}_{\gamma\delta} = \hat{C}_\gamma|_\delta + \hat{C}_\delta$.

For the first relation, taking $\gamma = \delta = S$ gives $\hat{C}_{S^2} = \hat{C}_S|_S + \hat{C}_S = r_f|_{(1+S)}$, and $S^2 = -I$, so Lemma 4.27 gives

$$r_f|_{(1+S)} = \hat{C}_{-I} = 0. \quad (76)$$

For the second, take $\gamma = T$, $\delta = S$, so that $U = TS$ and $\hat{C}_U = \hat{C}_T|_S + \hat{C}_S$. Applying the cocycle relation twice more,

$$\hat{C}_{U^3} = \hat{C}_U|_{U^2} + \hat{C}_{U^2} = \hat{C}_U|_{U^2} + \hat{C}_U|_U + \hat{C}_U = \hat{C}_U|_{(1+U+U^2)},$$

and $U^3 = -I$, so Lemma 4.27 makes the left side vanish. Substituting $\hat{C}_U = \hat{C}_T|_S + \hat{C}_S$ and using $\hat{C}_S = r_f$,

$$0 = \hat{C}_T|_{S(1+U+U^2)} + r_f|_{(1+U+U^2)}, \quad \text{so} \quad r_f|_{(1+U+U^2)} = -\hat{C}_T|_{S(1+U+U^2)}. \quad (77)$$

This is where the hypothesis $f \in S_k^!$ was used in Lemma 3.11, and it is the only place: there $\hat{C}_T$ vanished, here it need not.

The T -cochain is controlled by Φ alone, whatever path represents the T -segment. We isolate this, since it is used again below for segments that wind.

Lemma 4.29. *There is a constant A_0 , depending on k alone, such that for every $f \in F_k$ and every admissible $\hat{\gamma}_f^T$,*

$$\hat{C}_T = -(2\pi i)^{n+1} A_0 \int_{\hat{\gamma}_f^T} f(\tau) d\tau = -(2\pi i)^{n+1} A_0 \Phi(f). \quad (78)$$

In particular the right-hand side depends on the homotopy class of $\hat{\gamma}_f^T$ only through $\Phi(f)$, and Φ is by Definition 4.16 the integral over that same path.

Proof. The computation of $\hat{C}_T$ in the proof of Lemma 3.9 does not use $c_f(0) = 0$ until its final step. It shows that $A(\tau) := \chi(\tau; X + Y, Y) - \chi(\tau - 1; X, Y)$ is a 1-periodic polynomial, hence a constant A_0 depending on k alone, and leaves $\hat{C}_T$ as the stated multiple of $\int_{\hat{\gamma}_f^T} f d\tau$. No property of the path enters beyond its endpoints, which are fixed by Definition 2.1. $\square$

Since A_0 does not depend on f , neither does $\mathcal{D} := (2\pi i)^{n+1} A_0|_{S(1+U+U^2)}$, and (77) becomes

$$r_f|_{(1+U+U^2)} = \Phi(f) \mathcal{D}. \quad (79)$$

The defect is therefore proportional to $\Phi(f)$, with a constant of proportionality common to every $f \in F_k$ whose poles avoid $\mathrm{SL}_2(\mathbb{Z}) \cdot \gamma^{\mathrm{arc}}$.

It remains to identify $\mathcal{D}$. Apply (79) to $f = E_k$, which is holomorphic and so certainly admissible, and for which $\Phi(E_k) = 1$: this gives $\mathcal{D} = r_{E_k}|_{(1+U+U^2)}$. Hence

$$\hat{r}_f|_{(1+U+U^2)} = r_f|_{(1+U+U^2)} - \Phi(f) r_{E_k}|_{(1+U+U^2)} = \Phi(f) \mathcal{D} - \Phi(f) \mathcal{D} = 0,$$

while (76) applied to both f and E_k gives $\hat{r}_f|_{(1+S)} = 0 - \Phi(f) \cdot 0 = 0$. Both defining relations of W hold. Nothing in the argument constrains where the poles of f lie, or how deep they are, beyond their absence from $\mathrm{SL}_2(\mathbb{Z}) \cdot \gamma^{\mathrm{arc}}$, which was used only through Lemma 4.27. $\square$

Lemma 4.30. *Let $f \in F_k$ have poles on γ^{arc} , none at i and none on $\mathrm{SL}_2(\mathbb{Z}) \cdot \rho$, and let $\hat{\gamma}^S$ be S -symmetric. Write $D_E := r_{E_k}|_{(1+U+U^2)}$. Then $\hat{r}_f|_{(1+S)} = 0$ for every admissible $\hat{\gamma}^T$. If $\hat{\gamma}^T$ passes on the same side of every pole as $\hat{\gamma}^S$ then $\hat{r}_f|_{(1+U+U^2)} = 0$; otherwise, with $p_1, \dots, p_m$ the poles at which the two segments part,*

$$\hat{r}_f|_{(1+U+U^2)} = -(2\pi i)^{n+2} \sum_{j=1}^m \nu_j a_{p_j} (X - p_j Y)^n - 2\pi i \left(\sum_{j=1}^m X_T(p_j) a_{p_j} \right) D_E, \quad (80)$$

where $a_p = \mathrm{Res}_{\tau=p} f$ and $\nu_j \in \mathbb{Z} \setminus \{0\}$, and this is non-zero unless D_E lies in the span of $\{(X - p_j Y)^n\}_{j \leq m}$.

Proof. By Lemma 3.9, $\hat{C}_S = r_f$ with $\hat{C}_S = C_S - (E|_S - E)$ and $C_S = (2\pi i)^{n+1} \int_{\hat{\gamma}_S} \omega_f$; the T -segment enters r_f only through the coboundary $E - E|_S$. The cocycle relation gives $r_f|_{(1+S)} = \hat{C}_S|_{(1+S)} = \hat{C}_{S^2}$, and as n is even the coboundary drops, leaving the integral of ω_f over the closed path formed by $\hat{\gamma}^S$ followed by its S -image. In polar coordinates S acts by $(r, \theta) \mapsto (1/r, \pi - \theta)$, so an S -symmetric $\hat{\gamma}^S$ is carried onto itself with reversed orientation; that path retraces and the integral vanishes. No property of $\hat{\gamma}^T$ is used, and the subtracted term contributes nothing since $r_{E_k}|_{(1+S)} = 0$.

For the second relation, Φ is by construction the quantity $\hat{C}_T$ measures, so Definition 4.25 gives $\hat{C}_T = 0$; then $\hat{C}_U = \hat{C}_T|_S + \hat{C}_S = \hat{C}_S$ and $\hat{r}_f|_{(1+U+U^2)} = \hat{C}_{U^3} - \Phi(f) D_E$, with $\hat{C}_{U^3} = \hat{C}_{-I}$ the integral of ω_f over the closed path traced by $(TS)^3$. As $U = TS$, that path is assembled from

both segments. If $\hat{\gamma}^S = \hat{\gamma}^T$ it retraces, as in Lemma 4.27, and the two terms cancel exactly as in Lemma 4.28.

If the segments pass on opposite sides at $p_1, \dots, p_m$, the two paths bound regions containing exactly those poles and no others, and the residue theorem gives the first term of (80); the second is the shift induced on Φ , which is the T -segment integral and therefore moves with $\hat{\gamma}^T$. For the last claim, the coefficient vector of $(X - pY)^n$ is $((\binom{n}{l})(-p)^l)_l$, so the matrix of $\{(X - p_j Y)^n\}_{j \leq m}$ is $\text{diag}(\binom{n}{l})$ times a Vandermonde in $(-p_1, \dots, -p_m)$, of rank m whenever the p_j are distinct and $m \leq n + 1$. The residue terms therefore cannot cancel among themselves, and (80) can vanish only if D_E lies in their span. $\square$

The shifted contour realises one of the two sides without a second prescription being needed.

Lemma 4.31. *Let f be as in Lemma 4.30. Then for every $\epsilon > 0$ the pair $(\hat{\gamma}_\epsilon^S, \hat{\gamma}_\epsilon^T)$ of Definition 4.5 passes on the same side of every pole of f on γ^{arc} , and $\hat{r}_f$ computed on it is independent of ϵ and equals the value of Lemma 4.30 on that pair.*

Proof. The two segments differ only inside $|\tau - \rho| = O(\epsilon)$, where by hypothesis f has no pole, so they pass on the same side of each pole on the arc and Lemma 4.30 applies with $m = 0$ in (80). Independence of ϵ is Lemma 4.3: varying ϵ slides τ_0 radially without crossing a pole, the poles on the arc lying at $|\tau| = 1$ and the sides taken by (51) being the same for every $\epsilon > 0$. $\square$

The two relations are therefore sensitive to the two segments in quite different ways. The S -relation constrains $\hat{\gamma}^S$ alone: S -symmetry makes the S^2 -loop degenerate whatever $\hat{\gamma}^T$ does. The $(1 + U + U^2)$ relation mixes them, since $U = TS$, and vanishes only when the two segments agree. Requiring $\hat{\gamma}^S = \hat{\gamma}^T$ in Definition 4.4 is thus not a stipulation of convenience: membership in W selects it.

Lemma 4.32. *Let $f \in F_k$ have a pole on $\text{SL}_2(\mathbb{Z}) \cdot \rho$ and let $W = W(f, \epsilon)$ be as in Definition 4.6. Then $W \equiv -1 \pmod{3}$; every such value is realised by some connector; and there is a vector $c \in V_n$, independent of W , with*

$$\hat{r}_f|_{(1+S)} = 0, \quad \hat{r}_f|_{(1+U+U^2)} = -cW. \quad (81)$$

Proof. In the coordinate w of Definition 4.6 the map $U = TS$ fixes $w = 0$ and is exactly the rotation $w \mapsto \omega^{-1}w$, $\omega = e^{2\pi i/3}$, since $U'(\rho + 1) = (\rho + 1)^{-2} = e^{-2\pi i/3}$; winding about $\rho + 1$ is winding about $w = 0$, and U preserves increments of $\arg w$ exactly. The relator path is $\gamma_{U^3} = \gamma_U \cdot (U\gamma_U) \cdot (U^2\gamma_U)$, closed as $U^3 = -I$. Its three pieces are U -translates of one another, hence each changes $\arg w$ by the same amount Δ , and the endpoints $\tau_0(\epsilon), U\tau_0(\epsilon), U^2\tau_0(\epsilon)$ sit at arguments $\frac{\pi}{3}, -\frac{\pi}{3}, -\pi$, so $\Delta = -\frac{2\pi}{3} + 2\pi k$ with $k \in \mathbb{Z}$ the class of the connector. Then $2\pi W = 3\Delta$ gives $W = -1 + 3k$, and every k occurs.

For (81), the S -relation holds because $\hat{\gamma}_\epsilon^S$ is S -symmetric, so the S^2 -path retraces by Lemma 4.26 and $\hat{C}_{S^2} = 0$ whatever $\hat{\gamma}_\epsilon^T$ does; as $r_{E_k}|_{(1+S)} = 0$ the subtraction contributes nothing. For the U -relation, (77) gives $r_f|_{(1+U+U^2)} = -\hat{C}_T|_{S(1+U+U^2)} - \hat{C}_{U^3}$, and Lemma 4.29 holds at every winding, so the first term is $\Phi(f)\mathcal{D}$ with $\mathcal{D} = D_E$ exactly as in Lemma 4.28; Definition 4.25 cancels it for every W . There remains $\hat{C}_{U^3}$, which by Cauchy's theorem is $2\pi i \sum_q W_q \text{Res}_q \omega_f$ over the poles q of f . At $\epsilon = 0$ with equal segments the relator path retraces by Lemma 4.26; for $\epsilon \neq 0$ the two segments of Definition 4.5 differ only inside $|\tau - \rho| = O(\epsilon)$, so the retracing portions still cancel as chains and γ_{U^3} is homologous to a cycle supported in the six discs $U^j(D_{O(\epsilon)}(\rho + 1) \cup D_{O(\epsilon)}(\rho + 2))$, $j = 0, 1, 2$, whose centres all lie in $\text{SL}_2(\mathbb{Z}) \cdot \rho$. The poles of f are discrete and $\text{SL}_2(\mathbb{Z})$ -stable, hence locally finite, so for small ϵ those discs contain no pole but their centres and $W_q = 0$ for every

$q \notin \mathrm{SL}_2(\mathbb{Z}) \cdot \rho$. The cycle is invariant under U up to cyclic reordering, so W_q is constant on U -orbits and equals W on $\mathrm{SL}_2(\mathbb{Z}) \cdot \rho$; collecting the residues into c gives (81). $\square$

Corollary 4.33. *For f as in Lemma 4.32, $\hat{r}_f \in W$ in the sense of Definition 4.7, and that class is the unique one in which it does.*

Proof. By (81) the defect vanishes exactly at $W = 0$, and $c \neq 0$ since the defect is non-zero at $W = -1$; uniqueness follows because $W \mapsto -cW$ is injective. The same argument at i , where U is replaced by S and the relator path is the S^2 -path, gives W odd and the prescription the mean of the two indentations. $\square$

Lemmas 4.30 and 4.32 are complementary rather than overlapping: the first excludes i and $\mathrm{SL}_2(\mathbb{Z}) \cdot \rho$ by hypothesis, the second treats only those points, where the stabiliser forces W into a class that misses 0 and the L-integral must be read off the affine family rather than from a single contour. Together with Lemma 4.28 they cover every pole configuration.

Theorem 4.34. *For $f \in F_k$, and $W \subset V_n$ from Definition 3.10, the period polynomial $\hat{r}_f(X, Y) \in W$.*

Proof. Let $E \subset f^{-1}(\infty)$ be the poles of f lying on $\mathrm{SL}_2(\mathbb{Z}) \cdot \rho$ or on $\mathrm{SL}_2(\mathbb{Z}) \cdot i$, and N those lying on γ^{arc} but not at an elliptic point; the remaining poles meet γ^{arc} nowhere. The three mechanisms are local to these three families and may be applied together.

If E and N are both empty, Lemma 4.28 gives $\hat{r}_f \in W$ directly, with no condition on where the other poles lie or how deep they are.

If E is empty and N is not, Definition 4.4 takes $\hat{\gamma}_f^S = \hat{\gamma}_f^T$, and both relations hold by Lemma 4.30; the shifted contour of Definition 4.5 gives the same value by Lemma 4.31, so the two descriptions do not compete.

If E is non-empty, read L^* and $\hat{r}_f$ off the shifted class of Definition 4.5 at $W = 0$, as in Definition 4.7, at each elliptic point of E in turn. Lemma 4.32 shows the two defining relations vanish there, and Corollary 4.33 that $W = 0$ is the only class in which they do. Poles of f elsewhere are untouched and fall under the two previous cases. Hence $\hat{r}_f \in W$. $\square$

The construction uses only the $k - 1$ special values $L^*(\cdot, \ell + 1)$, $0 \leq \ell \leq n$, of the finite-contour L -function of Section 4.1, together with the single subtraction $\Phi(f) r_{E_k}$. No reference family is chosen, and no projection onto W is performed.

Two choices enter, and neither moves $\hat{r}_f$ out of W . In case 2 the side of the indentation is free, and by Lemma 4.30 the two differ within W . In case 3 nothing is free: Corollary 4.33 singles out $W = 0$, and ϵ is inert by Lemma 4.3. In Definition 4.25 the reference form E_k may be replaced by any g with $\Phi(g) = 1$, and two such choices shift $\hat{r}_f$ by $\Phi(f) r_{g-g'}$, where $g - g'$ runs over the forms annihilated by Φ ; how large that shift space is depends on the class from which g is drawn, and we do not settle it here.

A Meromorphic modular forms

For $f \in F_k$ with poles in $\mathbb{H}$, the two segment integrals of Definition 2.3 depend on their contours; we show their sum depends only on the homotopy class of the pair.

Proposition A.1. *For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, the L -integral $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends only on the homotopy class of its two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, and not on the basepoint τ_0 . At fixed τ_0 each of L_S^*, L_T^* is separately path-independent.*

Proof. At fixed τ_0 the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so Cauchy's theorem makes each of L_S^*, L_T^* constant on the homotopy class of its own segment. Under a motion of the basepoint, $\partial_{\tau_0} L^*$ is evaluated at the moving endpoints alone; the computation of Proposition 2.6, which uses only $f(\tau_0 - 1) = f(\tau_0)$ and $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, cancels the boundary terms of L_S^* against those of L_T^* , the interior poles lying off the contours. Hence $\partial_{\tau_0} L^* = 0$ along any motion in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, while each summand may separately vary. $\square$

Definition A.2. For $f \in F_k$ and $\tau_0 \in \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, let $[\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]$ denote the homotopy classes of the two segments in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Thus, with $\gamma_{\tau_0}^S \in [\gamma_{\tau_0}^S]$ and $\gamma_{\tau_0}^T \in [\gamma_{\tau_0}^T]$, by Proposition A.1 the value

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) := L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_S^*(f, s; \gamma_{\tau_0}^S) + L_T^*(f, s; \gamma_{\tau_0}^T) \quad (82)$$

is independent of the chosen representative paths and independent of τ_0 (subject to natural constraints of τ_0 monodromies around singular points, which move between homotopy classes).

We would like to speak of “the” L-integral. Homotopy classes of course split this into L-integrals for contours in different homotopy classes. And, so, in what follows, we define the reference homotopy classes:

Definition A.3 (Reference homotopy class for $f \in F_k$). Let $f \in F_k$, let τ_0 be the base-point of $L^*(f, s)$, and let $f^{-1}(\infty)$ be the poles of f . If f has no poles on the line $\text{Im}(\tau) = 1$, choose $\delta = 0$. Else, choose $\delta > 0$ such that no pole lies in the strip $1 < \text{Im}(\tau) \leq 1 + \delta$. Set $\hat{\tau}_0^f := i(1 + \delta)$. The reference homotopy class $[\hat{\gamma}_f] = [\hat{\gamma}_f^S] + [\hat{\gamma}_f^T]$ is:

- $[\hat{\gamma}_f^T]$: contains the horizontal segment from $T^{-1}\hat{\tau}_0^f$ to $\hat{\tau}_0^f$;
- $[\hat{\gamma}_f^S]$: contains the near-vertical path from $S^{-1}\hat{\tau}_0^f$ to $\hat{\tau}_0^f$ along the imaginary axis, taking a pole at i by principal value.

For $\delta = 0$, the reference paths match the paths $\gamma_i^S : i \rightarrow i$ and $\gamma_i^T : (i-1) \rightarrow i$ for $L^*(f, s)$ when $f \in M_k^1$.

With this definition, we may thus unambiguously write

$$L^*(f, s) := L^*(f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) . \quad (83)$$

Now, having fixed the reference value $L^*(f, s)$ in the reference homotopy class of Definition A.3, consider two paths $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ which may be in any homotopy class. Further, let $X_S(p), X_T(p) \in \mathbb{Z}$ count the windings $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ has about each pole $\tau_p \in f^{-1}(\infty)$ relative to the reference-class.

Lemma A.4. Let $f \in F_k$, $r_S(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tau^{s-1})$ and $r_T(f, \tau_p) := \text{Res}_{\tau=\tau_p}(f(\tau)\tilde{k}_T(\tau, s))$. Then,

$$L^*(f, s; [\gamma_{\tau_0}^S], [\gamma_{\tau_0}^T]) = L^*(f, s) + 2\pi i e^{-\pi i s/2} \sum_{\tau_p \in f^{-1}(\infty)} \left[X_S(\tau_p) r_S(f, \tau_p) + X_T(\tau_p) r_T(f, \tau_p) \right]. \quad (84)$$

Proof. Two pairs whose winding data differ by one unit at a single pole p , on one segment, bound a loop encircling p once; the residue theorem adds $2\pi i$ times the residue of that segment's integrand, $f(\tau)\tau^{s-1}$ for γ^S or $f(\tau)\tilde{k}_T(\tau, s)$ for γ^T , and the prefactor $e^{-\pi i s/2}$ of Definition 2.3 multiplies through. Summing over crossings gives (84). $\square$

This content provides the basics to discuss our finite L-functions for $f \in F_k$. However, we wish to present these L-functions in terms of their q-series coefficients, just as we have done for L-functions for $f \in M_k^!$ in the previous section.

The obstruction to doing so directly is not a technical one. For $f \in M_k^!$ the series $\sum_n c_f(n)q^n$ converges on all of $\mathbb{H}$, as f is holomorphic there, and the term-by-term evaluation of §3 is legitimate throughout. A pole at finite height ends this. The expansion about the cusp then represents f only above the highest pole, on $\text{Im } \tau > y_f := \sup\{\text{Im } \tau_p\}$, with coefficients growing as $|c_f(n)| \gtrsim n^{m-1} e^{2\pi n \text{Im } \tau_p}$ for a pole of order m [3]. The reference contour of Definition A.3 runs along $\text{Im } \tau = 1 + \delta$ (with $\delta = 0$ if f has no poles with $\text{Im}(\tau_p) = 1$), so as soon as f has a pole at or above that height the q -series diverges at every point of the segment. There is then no series to integrate term by term, rather than a slowly convergent one to be handled with care. To do this, we must introduce polylogarithmic projections of Ref. [3].

A.1 Explicit formulae for $L^*(f, s)$ for general $f \in F_k$

The polylog projection isolates the q -series contribution of a given pole of an $f \in F_k$ into a known special function, an integral of a polylogarithm of negative index, and leaves behind an object $(I - \hat{P})f$ with that one pole removed. Since the q -series of f diverges on the reference contour as soon as a pole sits at or above $\text{Im } \tau = 1$, it is essential for our purposes to project out every such pole: only then does what remains admit a q -series there.

Within this general context, we lay-out the starting definitions and facts:

Definition A.5. $F(\infty)$ is the space of meromorphic functions $g : \mathbb{H} \rightarrow \mathbb{C}$ with $g(\tau + 1) = g(\tau)$, finitely many poles in each compact subset of the strip $|\text{Re } \tau| \leq \frac{1}{2}$, and bounded pole order at $q = 0$.

Definition A.6. Let $f \in F_k$ have a pole of order m_p at $\tau_p \in \mathbb{H}$. On a punctured ε -ball about τ_p small enough to exclude the other poles of f , we have

$$f(\tau) = \sum_{m=1}^{m_p} \frac{r_{f,\tau_p}^*(m)}{(\tau - \tau_p)^m} + g(\tau), \quad r_{f,\tau_p}^*(m) = \frac{1}{2\pi i} \oint_{|\tau - \tau_p|=\varepsilon} f(\tau) (\tau - \tau_p)^{m-1} d\tau, \quad (85)$$

with g holomorphic at τ_p . We call $r_{f,\tau_p}^*(1), \dots, r_{f,\tau_p}^*(m_p)$ the pole weights of f at τ_p ; the first is the residue, and the last is nonzero, which is what fixes the order. We may then define the polylog projection of f at τ_p to be

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m=1}^{m_p} r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)}) , \text{ where } \text{Li}_s(x) := \sum_{n \geq 1} \frac{x^n}{n^s} . \quad (86)$$

Definition A.7. For every $f \in F_k$, there are a finite set of T -inequivalent finite order poles $\{\tau_p\}$ of order $\{m_p\}$. We may use the polylogarithm projector of Definition A.6 to define two functions $\hat{P}(f)$ and $(1 - \hat{P})(f) \in F(\infty)$, via the composite projection operator:

$$\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p} . \quad (87)$$

Their respective singularity structures are integral to our explicit L-function expressions, below.

Lemma A.8. The normalized block $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has principal part exactly $(\tau - \tau_p)^{-m}$, so $\hat{P}_{\tau_p} f$ matches the principal part of f and $(I - \hat{P}_{\tau_p})f$ is holomorphic at τ_p .

Proof. By the small- t expansion $\text{Li}_{1-m}(e^{-t}) = (m-1)!t^{-m} + O(t^0)$ with $t = -2\pi i(\tau - \tau_p)$, the block $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ has a *pure* pole $(\tau - \tau_p)^{-m} + O((\tau - \tau_p)^0)$ at τ_p , with no lower-order poles. Summing against the Laurent coefficients $r_{f, \tau_p}^*(m)$ therefore reproduces the principal part of f term by term, leaving $(I - \hat{P}_{\tau_p})f$ regular at τ_p . $\square$

Definition A.9. Let $f \in F_k$ and $\hat{P}$ as in Definition A.7. We denote the Fourier coefficients of $(I - \hat{P})f$ as follows:

$$(I - \hat{P})f(\tau) = \sum_n c_{Rf}(n) q^n, \quad q = e^{2\pi i \tau}. \quad (88)$$

Lemma A.10. $F_k \subset F(\infty)$ and $(I - \hat{P})f \in F(\infty)$. Every pole of $(I - \hat{P})f$ has $\text{Im } \tau < 1$, with a maximal height $y_b < 1$ (set $y_b = 0$ if $(I - \hat{P})f$ is holomorphic).

Proof. A weakly holomorphic form is T -periodic, meromorphic, and $O(e^{C\text{Im } \tau})$ at the cusp, so $F_k \subset F(\infty)$; the projected polylogs $\text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ share these properties, so $(I - \hat{P})f \in F(\infty)$ (and leaves F_k , being non-modular). By Lemma A.8, $\hat{P}$ cancels the principal part of every pole with $\text{Im } \tau \geq 1$, so the poles of $(I - \hat{P})f$ lie strictly below $\text{Im } \tau = 1$; finite in each compact set and accumulating only at $\text{Im } \tau = 0$, they have a maximal height $y_b < 1$. $\square$

The Fourier coefficients of a T -periodic meromorphic function are controlled by its highest pole: $|c_{Rf}(n)| \leq e^{2\pi n(y_b + \varepsilon)}$ for every $\varepsilon > 0$ and all large n [3]. Since $y_b < 1$, the series $\sum_n c_{Rf}(n) q^n$ converges geometrically down to $\text{Im } \tau = 1$, at base $e^{-2\pi(1 - y_b - \varepsilon)}$, and so on $\hat{\gamma}_f^T$ itself. This is the point of the projection: convergence merely for large $\text{Im } \tau$ would be automatic and of no use, as the contour sits at height one.

For $n \neq 0$, define

$$G_{s,k}(n) := \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k - s, 2\pi n)}{(2\pi n)^{k-s}}. \quad (89)$$

Each pole with $\text{Im } \tau_p \geq 1$ contributes one projected polylog to $L^*(f, s)$, whose L-integral we now record.

Definition A.11. For a pole of order $N + 1$ projected at z with $\text{Im } z > 1$, let $I_N(s, z)$ be the L-integral of the projected polylog $\lambda_{N,z}$ in the minimal class $[\hat{\gamma}_f^S], [\hat{\gamma}_f^T]$ of Definition A.3:

$$I_N(s, z) := L^*(\lambda_{N,z}, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]), \quad (90)$$

where $\lambda_{N,z}$ denotes the function $\text{Li}_{-N}(e^{2\pi i(\tau - z)})$, fed into the linear L^* -functional.

Lemma A.12. Let $f \in F_k$ have a pole of order p at z , $\text{Im } z > 1$, with projection weights $r_{f,z}^*(m)$ (Definition A.6). Polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w) - \delta_{N,0}$ ($N \in \mathbb{N}_0$) resums (90) into (each exponentially convergent, base $e^{2\pi(1 - \text{Im}(z))}$)

$$L^*(\hat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (91)$$

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} G_{s,k}(-n). \quad (92)$$

Proof. By Proposition A.1 we may evaluate $I_N(s, z)$ in the $\delta \rightarrow 0$ representative of the minimal class; there the projected polylog, which is regular at i since $\text{Im } z > 1$, has vanishing γ^S -integral and $\hat{\gamma}_f^T \rightarrow [i-1, i]$. On $[i-1, i]$ one has $|e^{2\pi i(\tau-z)}| > 1$, so the defining series of Li_{-N} diverges; inversion rewrites it as $(-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n(z-\tau)}$, convergent since $|e^{2\pi i(z-\tau)}| < 1$, and term-by-term integration against $\tilde{k}_T$ gives $\sum_n n^N e^{2\pi i n z} G_{s,k}(-n)$, the mode- $(-n)$ block of Theorem 3.2; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ contributes $\frac{1}{s} + \frac{i^k}{k-s}$. The rate is $e^{-2\pi(\text{Im } z-1)}$, from $\Gamma(s, -2\pi n) \sim (2\pi n)^{s-1} e^{2\pi n}$ against $|e^{2\pi i n z}| = e^{-2\pi n \text{Im } z}$. $\square$

Write $(a)_{(\ell)} := a(a-1)\cdots(a-\ell+1)$ for the falling factorial, with $(a)_{(0)} := 1$. For any integer $J \geq 1$, the incomplete gamma satisfies

$$\Gamma(s, X) = e^{-X} \sum_{\ell=0}^{J-1} (s-1)_{(\ell)} X^{s-1-\ell} + (s-1)_{(J)} \Gamma(s-J, X). \quad (93)$$

It peels the leading J terms of the large- X expansion of $\Gamma(s, X)$ into elementary powers $X^{s-1-\ell}$ and collects the rest into an incomplete gamma of lower order $s-J$ and hence faster-decaying. Importantly, the peeled powers integrate and resum against our kernel in closed form. Choosing J large enough therefore quarantines the entire singular structure at $\text{Im}(z) = 1$ into that finite, explicit sum, while the remainder's mode-sum is left manifestly convergent. We use (93) first for the poles on the line $\text{Im}(z) = 1$ with $z \neq i$, and then precisely at $z = i$.

Extracting similar results for L^* evaluated on the polar terms for poles on the line $\text{Im}(\tau) = 1$ requires more care. To do this away from the special point $z = i$, we find it helpful to define the following two functions:

$$\Pi_N(s, z) := \sum_{\ell=0}^N \frac{\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\ell+1}} \{(s-1)_{(\ell)} + i^k(k-s-1)_{(\ell)}\}, \quad (94)$$

$$R_N(s, z) := \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left[(s-1)_{(N+1)} \frac{\Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} + i^k(k-s-1)_{(N+1)} \frac{\Gamma(k-s-N-1, -2\pi n)}{(-2\pi n)^{k-s}} \right], \quad (95)$$

with Li_s as in Definition A.6, and $\Gamma(a, x)$ the upper incomplete gamma taken on its principal branch [10].

Lemma A.13. *Let $f \in F_k$ have a pole of order p at z with projection weights $r_{f,z}^*(m)$ (Definition A.6). With Π_N, R_N as in (94)–(95), for $\text{Im } z > 1$ and every $s \in \mathbb{C}$,*

$$L^*(\hat{P}_z f, s; [\hat{\gamma}_f^S], [\hat{\gamma}_f^T]) = \sum_{m=1}^p r_{f,z}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, z), \quad (96)$$

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} [\Pi_N(s, z) + R_N(s, z)], \quad (97)$$

and the second line's right side extends $I_N(s, \cdot)$ continuously to $\text{Im } z = 1$ for $z \notin i + \mathbb{Z}$.

The crux of this Lemma is to show the existence of the smooth continuation of $I_N(s, z)$ from Lemma A.12, derived for $\text{Im}(z) > 1$, to the $\text{Im}(z) = 1$ -line away from the elliptic point $z = i$. This elliptic point requires special treatment and care, involving specific relations between Laurent coefficients at the special elliptic point $\tau = i$. We shall deal with this special case separately.

Proof. We manipulate the main result of Lemma A.12,

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n=1}^{\infty} n^N e^{2\pi i n z} \left(\frac{\Gamma(s, -2\pi n)}{(-2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi n)}{(-2\pi n)^{k-s}} \right), \quad (98)$$

Our proof proceeds in two steps. First, we isolate the possible sources of divergence on the line $\text{Im}(z) = 1$ into a finite sum of polylogarithms $\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})$ for $\ell = 0, \dots, N$. To do this, we apply the recursion (93) to each gamma in $G_{s,k}(-n)$ at $X = -2\pi n$, $J = N+1$. Dividing each incomplete-Gamma term by $(-2\pi n)^s$ and $(-2\pi n)^{k-s}$ yields, then, a common factor of $1/(-2\pi n)^{\ell+1}$ for each term in the finite sum. Further, this finite-sum is proportional to $e^{2\pi n}$; we leave the trailing incomplete gammas as-is:

$$G_{s,k}(-n) = \left\{ e^{2\pi n} \sum_{\ell=0}^N \frac{(s-1)_{(\ell)}}{(-2\pi n)^{\ell+1}} + \frac{(s-1)_{(N+1)} \Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} \right\} + i^k (s \rightarrow k-s). \quad (99)$$

Now, we consider the n -sum $G_{s,k}(-n)$, starting with the falling factorial terms:

$$\sum_{n=1}^{\infty} n^N e^{2\pi i n z} e^{2\pi n} \left(\sum_{\ell=0}^N \frac{(s-1)_{(\ell)}}{(-2\pi n)^{\ell+1}} + i^k (s \rightarrow k-s) \right) = \sum_{\ell=0}^N \frac{\text{Li}_{\ell+1-N}(e^{2\pi i(z-i)})}{(-2\pi)^{\ell+1}} \{ (s-1)_{(\ell)} + i^k (k-s-1)_{(\ell)} \}, \quad (100)$$

i.e., $\Pi_N(s, z)$ of (94), while the reduced-gamma terms remaining are $R_N(s, z)$ of Eq. (95), which together establish the lemma's identity (97).

The second main step of our proof is to show that $\Pi_N(s, z)$ is regular as $\text{Im}(z) \rightarrow 1$ for $z \neq i$, while $R_N(s, z)$ is regular for all z with $\text{Im}(z) = 1$. We treat each piece in turn, beginning with R_N .

Each term of R_N is internal to the $G_{s,k}(-n)$ in the mode $\sum e^{2\pi i n z} G_{s,k}(-n)$ from Lemma A.12. Clearly, $|e^{2\pi i n z}| = e^{-2\pi n \text{Im} z}$. Thus, the incomplete gamma's exponential growth cancels against the exponentially falling $e^{2\pi i n z}$ factors. As $n \rightarrow \infty$, we have:

$$\left| n^N e^{2\pi i n z} \frac{\Gamma(s-N-1, -2\pi n)}{(-2\pi n)^s} \right| \sim \left| \frac{n^N e^{2\pi i n z} (-2\pi n)^{s-N-2} e^{2\pi n}}{(-2\pi n)^s} \right| \sim \left| \frac{e^{2\pi i n z} e^{2\pi n}}{(-2\pi)^{2+N} n^2} \right| \sim \frac{e^{2\pi n(1-\text{Im}(z))}}{(2\pi)^{N+2} n^2} \quad (101)$$

where the detailed asymptotics of $\Gamma(s, x)$ in the first scaling relation follow from [10, §8.11]. (The $i^k(s \rightarrow k-s)$ -partner follows identically.) Hence the n -th term of R_N is $O(n^{-2} e^{-2\pi n(\text{Im} z - 1)})$, uniformly for $\text{Im} z \geq 1$. There $e^{-2\pi(\text{Im} z - 1)} \leq 1$, so $\sum_n n^{-2} e^{-2\pi n(\text{Im} z - 1)} \leq \zeta(2)$: the series converges absolutely and uniformly, up to and including the edge $\text{Im} z = 1$, so R_N is continuous there.

For Π_N , recall that it is a sum over $N+1$ polylogarithms of integer index ≤ 1 , $\Pi_N(s, z) = c_1(s) \text{Li}_1(w) + c_0(s) \text{Li}_0(w) + \dots + c_{1-N}(s) \text{Li}_{1-N}(w)$, where $w := e^{2\pi i(z-i)}$. Each Li_a ($a \leq 1$) is single-valued and analytic on the cut plane $\mathbb{C} \setminus [1, \infty)$: for $a \leq 0$ it is rational with a pole of order $1-a$ at $w = 1$, while $\text{Li}_1(w) = -\log(1-w)$ has a logarithmic branch point there. Every singularity sits at the endpoint $w = 1$, and the cut of Li_1 runs along the ray $[1, \infty)$. The edge is the unit circle $|w| = 1$, which meets that radial cut only at $w = 1$; approaching it from inside the disc ($\text{Im} z > 1$) gives the analytic boundary value of each Li_a at every $w \neq 1$. Hence Π_N is analytic along the whole edge except at $z \in i + \mathbb{Z}$, where each term diverges. With the continuity of R_N , (97) seamlessly extends $I_N(s, \cdot)$ continuously to $\text{Im} z = 1$ for $z \notin i + \mathbb{Z}$. $\square$

The above result is insufficient to treat with the Laurent series singularity patterns of a modular form $f \in F_k$ which has poles at $\tau = i$, where Π_N is singular ($w = 1$). However, as we will

presently show, even though each individual polylogarithm of index at most one diverges at the point $z = i$, the special elliptic nature of that point imposes constraints on the sum of naively divergent terms such that the divergences exactly cancel. Before doing this, we find it helpful to present an important definition, and an associated remark:

Definition A.14. For $N \geq 0$, with $H_N := \sum_{r=1}^N \frac{1}{r}$ and $c_\ell^+(s) := (s-1)_{(\ell)} + i^k(k-s-1)_{(\ell)}$, define:

$$\mathcal{E}_N^+(s) := \sum_{n \geq 1} n^N e^{2\pi n} \left[(s-1)_{(N+1)} \frac{\Gamma(s-N-1, 2\pi n)}{(2\pi n)^s} + i^k(k-s-1)_{(N+1)} \frac{\Gamma(k-s-N-1, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (102)$$

$$\text{FP}_N(s, k) := \frac{c_N^+(s) H_N}{(2\pi)^{N+1}} + \sum_{\ell=0}^{N-1} \frac{c_\ell^+(s) \zeta(\ell+1-N)}{(2\pi)^{\ell+1}} + \frac{(-1)^N N!}{(2\pi)^{N+1}} \sum_{t=0}^{N-1} \frac{(-1)^t (s-1)_{(t)}}{t! (N-t)} + \mathcal{E}_N^+(s). \quad (103)$$

Note: $\mathcal{E}_N^+(s)$ and $R_N(s, i)$ of (95) resemble one another because both come from expanding $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ and integrating term by term; they are not equal, differing by the inversion $e^{2\pi i(\tau-z)} \mapsto e^{-2\pi i(\tau-z)}$. On the contour that argument has modulus $e^{2\pi(\text{Im } z - 1)}$: for $\text{Im } z > 1$ the un-inverted series diverges, so Lemma A.12 inverts and R_N inherits the arguments $-2\pi n$, whereas at $z = i$ the modulus is 1, inversion buys nothing, and $\mathcal{E}_N^+$ resums the un-inverted series with $+2\pi n$.

Remark A.15. The remainders R_N of Lemma A.13 (95) and $\mathcal{E}_N^+$ of Definition A.14 (102) are what is left after unfolding J terms of the incomplete-gamma recursion (93); both use $J = N + 1$. The unfolding isolates the terms that could diverge as $\text{Im}(z) \rightarrow 1^+$ into an explicit finite set of special functions, and $J = N + 1$ is the least J that isolates all of them, and leaves a remainder that converges for every z on $\text{Im}(z) = 1$. Any larger J isolates the same terms, leaving I_N and FP_N unchanged and merely quickening the remainder's decay ($\sim n^{-(J-N+1)}$) at the cost of carrying further positive-index polylogs $\text{Li}_2, \dots, \text{Li}_{J-N}$ into the finite sum (contributing values up to $\zeta(J-N)$ at $z = i$).

Lemma A.16. Let $f \in F_k$ have a pole of order P at i , with pole weights $r_{f,i}^*(m)$ (Definition A.6), i.e. principal part $\sum_{m=1}^P r_{f,i}^*(m)(\tau - i)^{-m}$. Modularity $f(-1/\tau) = \tau^k f(\tau)$ is equivalent to one relation for each principal-part order μ ,

$$\sum_{m=\mu}^P r_{f,i}^*(m) \binom{m}{\mu} i^m = i^k \sum_{m=\mu}^P r_{f,i}^*(m) \binom{k}{m-\mu} i^{-m}, \quad \forall \mu = 1, \dots, P. \quad (104)$$

The leading relation ($\mu = P$) is $((-1)^P - i^k) r_{f,i}^*(P) = 0$, so the pole order matches $k \bmod 4$ (P odd $\Leftrightarrow k \equiv 2$, P even $\Leftrightarrow 4 \mid k$).

Note: at $P = 1$, the Laurent-series relations claimed in this Lemma work out to be this is $(1 + i^k) r_{f,i}^*(1) = 0$; for a double pole the $\mu = 1$ relations are $r_{f,i}^*(1) = \frac{i(k-2)}{2} r_{f,i}^*(2)$.

Proof. Write $\tau = i + \varepsilon$. Then $-1/\tau - i = -\varepsilon/(1 - i\varepsilon)$, so the principal part of $f(-1/\tau)$ is $\sum_m r_{f,i}^*(m)(i - \varepsilon^{-1})^m$ and that of $\tau^k f(\tau)$ is $i^k(1 - i\varepsilon)^k \sum_m r_{f,i}^*(m)\varepsilon^{-m}$. Matching the coefficient of $\varepsilon^{-\mu}$ through $(i - \varepsilon^{-1})^m = \sum_j \binom{m}{j} i^{m-j} (-1)^j \varepsilon^{-j}$ and $(1 - i\varepsilon)^k = \sum_l \binom{k}{l} (-i)^l \varepsilon^l$ gives $(-1)^\mu \sum_{m=\mu}^P r_{f,i}^*(m) \binom{m}{\mu} i^{m-\mu} = i^k \sum_{m=\mu}^P r_{f,i}^*(m) \binom{k}{m-\mu} (-i)^{m-\mu}$; dividing through by i^μ (with $-1 = i^2$ and $-i = i^{-1}$) yields (104). $\square$

Lemma A.17. *Let $f \in F_k$ have a pole of order P at i with pole weights $r_{f,i}^*(m)$ (Definition A.6), and $\hat{P}_i f$ its projection. Evaluated in the minimal class at basepoint $i(1+\delta)$, the pole's contribution to $L^*(f, s)$ has the finite $\delta \rightarrow 0$ limit*

$$L^*(\hat{P}_i f, s) := \lim_{\delta \rightarrow 0} L^*(\hat{P}_i f, s; i(1+\delta)) = \sum_{m=1}^P \frac{(-2\pi i)^m r_{f,i}^*(m)}{(m-1)!} \text{FP}_{m-1}(s, k), \quad (105)$$

entire in s . The limit exists precisely because the pole weights obey the relations of Lemma A.16 at the S -fixed point i : they cancel the individually divergent contour pieces.

Proof. We begin by noting that, at its branch point $w = e^{2\pi i(\tau-i)} \sim 1$ the polylogarithm expands as $\text{Li}_{1-m}(e^{2\pi i(\tau-i)}) = (m-1)!(-2\pi i(\tau-i))^{-m} + \sum_{p \geq 0} \frac{\zeta(1-m-p)}{p!} (2\pi i(\tau-i))^p$ ([10, §25.12]). Thus,

$$\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau-i)}) = (\tau-i)^{-m} + \frac{(-2\pi i)^m}{(m-1)!} \sum_{p=0}^{\infty} \frac{\zeta(1-m-p)}{p!} (2\pi i(\tau-i))^p, \quad (106)$$

is a pure order- m pole, as in Lemma A.8, plus a tail holomorphic on $|\tau-i| < 1$. Hence $\hat{P}_i f = \sum_{m=1}^P r_{f,i}^*(m)(\tau-i)^{-m} + (\text{regular at } i)$. As the contour $\text{Im}(\hat{\gamma}_f^T)$ has height $1+\delta > 1$, the analogous step of w -inversion, $\text{Li}_m(w) = -\delta_{m,0} - (-1)^m \text{Li}_m(1/w)$ from the proof of Lemma A.12 does not need to be invoked. Thus there is no $\frac{1}{s} + \frac{i^k}{k-s}$ in the final result.

We now extract their finite parts of this polylogarithm on the $\hat{\gamma}_f^T$ and $\hat{\gamma}_f^S$ contour-integrals.

On $\hat{\gamma}_f^T$ the block is the convergent series $\text{Li}_{1-m}(e^u) = \sum_{n \geq 1} n^{m-1} e^{2\pi i n \tau} e^{2\pi i n \tau}$; integrating against $\tilde{k}_T$ from the basepoint $\tau_0 = i(1+\delta)$ gives, by Theorem 3.2, $\sum_{n \geq 1} n^{m-1} e^{2\pi i n} \left[\frac{\Gamma(s, X_n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, X_n)}{(2\pi n)^{k-s}} \right]$ with $X_n := -2\pi i n \tau_0 = 2\pi n(1+\delta)$. Expanding by the recursion (93) at $J = m$, and using $e^{2\pi i n} e^{-X_n} = e^{-2\pi n \delta}$ and $X_n^{s-1-\ell} = (2\pi n)^{s-1-\ell} (1+\delta)^{s-1-\ell}$, the peeled sum over $0 \leq \ell \leq m-1$ becomes (with $c_\ell^+(s)$ collecting the s - and $i^k(k-s)$ -parts)

$$\sum_{\ell=0}^{m-1} \frac{c_\ell^+(s) (1+\delta)^{s-1-\ell}}{(2\pi)^{\ell+1}} \text{Li}_{\ell+2-m}(e^{-2\pi \delta}) + \mathcal{E}_{m-1}^+(s), \quad (107)$$

$\mathcal{E}_{m-1}^+$ being the reduced- Γ remainder of (102). With $\text{Li}_{\ell+2-m}(e^{-2\pi \delta}) = \Gamma(m-\ell-1)(2\pi \delta)^{\ell-m+1} + \sum_{p \geq 0} \frac{\zeta(\ell+2-m-p)}{p!} (-2\pi \delta)^p$, two finite (δ^0) parts survive:

1. The regular part at δ^0 gives, for $\ell \neq m-1$, the ζ -value $\frac{c_\ell^+(s)}{(2\pi)^{\ell+1}} \zeta(\ell+2-m)$: the ζ -sum of (103).
2. For $\ell < m-1$ the singular $\Gamma(m-\ell-1)(2\pi \delta)^{\ell-m+1}$ crosses the $\delta^{m-\ell-1}$ coefficient $\binom{s-1-\ell}{m-\ell-1}$ of $(1+\delta)^{s-1-\ell}$; by the falling-factorial identity $(s-1)_{(\ell)}(m-\ell-2)! \binom{s-1-\ell}{m-\ell-1} = (s-1)_{(m-1)}/(m-\ell-1)$ (and its $k-s$ partner) these sum to

$$\frac{c_{m-1}^+(s)}{(2\pi)^m} \sum_{\ell=0}^{m-2} \frac{1}{m-\ell-1} = \frac{c_{m-1}^+(s) H_{m-1}}{(2\pi)^m}, \quad (108)$$

the harmonic term of (103).

The remaining $\ell = m-1$ term is just $\log(2\pi \delta)$, which cancels in the $\delta \rightarrow 0$ limit of the S -contour. We now turn to this S -contour's contribution.

On $\hat{\gamma}_f^S$ the pure pole $\sum_m r_{f,i}^*(m)(\tau - i)^{-m}$, integrated by principal value against τ^{s-1} along the axis ($\tau = i(1 + v)$), running from τ_0 to $S\tau_0 = i/(1 + \delta)$, so that $-\delta/(1 + \delta) \leq v \leq \delta$, is elementary. Expanding $(1 + v)^{s-1} = \sum_t \frac{(s-1)(t)}{t!} v^t$, the terms $t \geq m$ are regular and $O(\delta)$, the term $t = m - 1$ integrates to $\log(1 + \delta) = O(\delta)$, and each $t \leq m - 2$ contributes

$$\text{PV} \int_{-\delta/(1+\delta)}^{\delta} \frac{v^t}{v^m} dv = \frac{\delta^{t-m+1}}{t-m+1} \left[1 - (-1)^{t-m+1} (1 + \delta)^{m-t-1} \right]. \quad (109)$$

Expanding $(1 + \delta)^{m-t-1}$, its terms δ^j with $j < m - t - 1$ carry the endpoint poles $\delta^{-(m-1)}, \dots, \delta^{-1}$, while the top term $j = m - t - 1$ leaves the δ^0 finite part

$$i^{-m} \text{PV} \int_{-\delta/(1+\delta)}^{\delta} \frac{(1 + v)^{s-1}}{v^m} dv \Big|_{\delta^0} = i^{-m} (-1)^{m-1} \sum_{t=0}^{m-2} \frac{(-1)^t (s-1)(t)}{t! (m-t-1)}, \quad (110)$$

the polynomial term of (103). Collecting, the finite part of block m is $\frac{(-2\pi i)^m}{(m-1)!} r_{f,i}^*(m) \text{FP}_{m-1}(s, k)$.

Divergences cancel by (104). The residual pieces are the $\delta \rightarrow 0$ divergences of the pole part $\sum_m r_{f,i}^*(m)(\tau - i)^{-m}$: the powers $\delta^{-(m-1)}, \dots, \delta^{-1}$, carried by the endpoints of $\hat{\gamma}_f^S$ straddling i as in (109) and by the $\ell < m - 1$ terms of (107), together with a $\log \delta$ carried by $\text{Li}_1(e^{-2\pi\delta})$ at $\ell = m - 1$ alone. At order μ the coefficient of the powers is

$$\sum_{m=\mu}^P r_{f,i}^*(m) \binom{m}{\mu} i^m - i^k \sum_{m=\mu}^P r_{f,i}^*(m) \binom{k}{m-\mu} i^{-m}, \quad (111)$$

precisely the difference of the two sides of the modularity relation (104), hence 0 for $f \in F_k$. The $\log \delta$ coefficient, summed over blocks with the normalisation $\frac{(-2\pi i)^m}{(m-1)!}$, is $-\sum_{m=1}^P (-i)^m r_{f,i}^*(m) \left[\binom{s-1}{m-1} + i^k \binom{k-s-1}{m-1} \right]$, which the same relations annihilate. Every $\delta^{-\mu}$ and $\log \delta$ term therefore cancels and the $\delta \rightarrow 0$ limit exists. Conversely, weights violating (104) at order μ leave that piece uncanceled (at $\mu = 1$, a $\log \delta$), so the limit exists only for $f \in F_k$.

The finite part and the (vanishing) divergences give (105). $\square$

Theorem A.18. *Let $f \in F_k$ have poles τ_p of order P_p with $\text{Im } \tau_p \geq 1$ in the strip $|\text{Re } \tau| \leq \frac{1}{2}$, pole-projected Fourier coefficients $c_{Rf}(n)$ (Definition A.9), and projection weights $r_{f,\tau_p}^*(m)$ (Definition A.6). In the minimal class,*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) G_{s,k}(n) + \sum_{\tau_p \neq i} \sum_{m=1}^{P_p} r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} I_{m-1}(s, \tau_p) + L^*(\hat{P}_i f, s), \quad (112)$$

with I_{m-1} from Lemmas A.12–A.13 and the last term (present only for a pole at i) from Lemma A.17. The right side is entire in s apart from at most simple poles at $s = 0$ and $s = k$, and reduces to the weakly-holomorphic L^* of Theorem 3.2 when f is holomorphic on $\mathbb{H}$.

Proof. Split $f = (I - \hat{P})f + \hat{P}f$. By Lemma A.10 the Fourier series of the pole-projected part converges on $\hat{\gamma}_f^T$; Theorem 3.2 applied term by term gives its L^* as the first two sums. The projection $\hat{P}f = \sum_{\tau_p} \sum_m r_{f,\tau_p}^*(m) \frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau - \tau_p)})$ contributes, for each $\tau_p \neq i$, the L-integrals $I_{m-1}(s, \tau_p)$ of Definition A.11, evaluated by Lemmas A.12–A.13; the pole at i contributes $L^*(\hat{P}_i f, s)$ of Lemma A.17. Entirety and the pole locations follow from those of $G_{s,k}$ and the $\delta_{N,0}$ terms; when f is holomorphic on $\mathbb{H}$, $\hat{P}f = 0$ and $c_{Rf} = c_f$, recovering Theorem 3.2. $\square$

A.2 Relations of this work to L-functions for $f \in F_k$ in [3]

We briefly pause to make two remarks that connect the L-functions for $f \in F_k$ defined in this note to the slightly different definition in Ref. [3].

Remark A.19. Casual inspection of the structure of the $I_N(s, \tau_p)$ -functions entering into Theorem A.18 would lead one to conclude that they substantively differ from those in [3], where the author first defined an L-function for $f \in F_k$: Ref. [3] has the $1/s$ -residue proportional only to $c_f(0)$. However, our L-function has $1/s$ -residue contributions from all *simple* poles located above the $\text{Im}(\tau_p) > 1$ -line. This naive difference is immaterial, and does not impact the results of the residue theorems for the $1/s$ -poles in [3], for the simple reason that in Ref. [3], the reference contour was implicitly at $\tau_0 \rightarrow i\infty$. This was valid in the restricted context of [3], as that paper only focused on $f \in F_k$ where $f = \sum_{n \geq 0} c_f(n)q^n$; poles at the $\text{SL}_2(\mathbb{Z})$ cusp were explicitly excluded, which allowed the limit $\tau_0 \rightarrow i\infty$ to be smooth and safe. However, in this work, we make a different reference contour — that in Definition A.3.

Remark A.20. The $1/s$ residue is carried entirely by the T -segment: L_S^* is a finite integral of $f \tau^{s-1}$ and is entire in s , while $\tilde{k}_T$ has a simple pole at $s = 0$ whose coefficient is $\int_{\hat{\gamma}_f^T} f d\tau$, the constant Fourier mode of f in the strip containing that segment. Deforming $\hat{\gamma}_f^T$ to pass above every pole of f therefore returns $\text{Res}_{s=0} L^*(f, s) = -c_f(0)$, the constant term at the cusp, and recovers the residue bookkeeping of [3]. That class differs from the one of Definition A.3 by the crossings of Lemma A.4.

A.3 Period polynomials for generic $f \in F_k$

In this section, we use our L-functions to assign period polynomials $r_f \in W \subset V_{k-2}$ to general elements $f \in F_k$. The route is straightforward when f has poles exclusively outside a certain region $\mathcal{T} \subset \mathbb{H}$, lying in the strip $0 \leq \text{Re } \tau \leq 1$ directly beneath the T -segment. We denote the set of such functions as $F_k^\circ \subset F_k$. For every $f \in F_k^\circ$ the period polynomial constructed from the special L-values of $L^*(f, s)$ of Theorem A.18 is immediately an element of W . However, when f has poles within $\mathcal{T}$, more care is needed. Ref. [15] gives a parallel construction of r_f for $f \in F_k$, which we re-present alongside one of our own. It is not obvious to us which construction should be preferred.

To begin, we define $\mathcal{T}$, then F_k° , and then r_f for $f \in F_k^\circ$:

Definition A.21. Let $\mathcal{T} \subset \mathbb{H}$ be the region

$$\mathcal{T} := \left\{ \tau \in \mathbb{H} : 0 \leq \text{Re } \tau \leq 1, \frac{1}{2} \leq \text{Im } \tau \leq 1, \left| \tau - \frac{i}{2} \right| \geq \frac{1}{2}, \left| \tau - 1 - \frac{i}{2} \right| \geq \frac{1}{2} \right\}. \quad (113)$$

$\mathcal{T}$ is the closed curvilinear triangle with vertices i , $1+i$ and $\frac{1+i}{2}$, bounded above by the line $\text{Im } \tau = 1$ and below by its $\text{SL}_2(\mathbb{Z})$ -translates through the cusps 0 and 1, the circles $|\tau - \frac{i}{2}| = \frac{1}{2}$ and $|\tau - 1 - \frac{i}{2}| = \frac{1}{2}$, which meet each other tangentially at $\frac{1+i}{2}$. Its three sides are the T -, UT - and U^2T -translates of $\hat{\gamma}_f^T$, meeting at those vertices; since $(TS)^3 = 1$ the last of these is S^{-1} , so $\partial\mathcal{T}$ is the U -orbit of $T\hat{\gamma}_f^T$. Accordingly U cycles $i \rightarrow 1+i \rightarrow \frac{1+i}{2} \rightarrow i$ and $\mathcal{T}$ is U -invariant. Note that $\hat{\gamma}_f^T$ runs from $i-1$ to i , so it is $T\hat{\gamma}_f^T$, not the segment itself, that caps $\mathcal{T}$; the region directly beneath the contour is $T^{-1}\mathcal{T}$. The bound $\text{Im } \tau \geq \frac{1}{2}$ is not redundant: dropping it leaves a second region below the point of tangency, widening to the full interval $0 \leq \text{Re } \tau \leq 1$ as $\text{Im } \tau \rightarrow 0$ and so of infinite hyperbolic area. Figure 1a draws $\mathcal{T}$, and locates it against the fundamental domain and its translates.

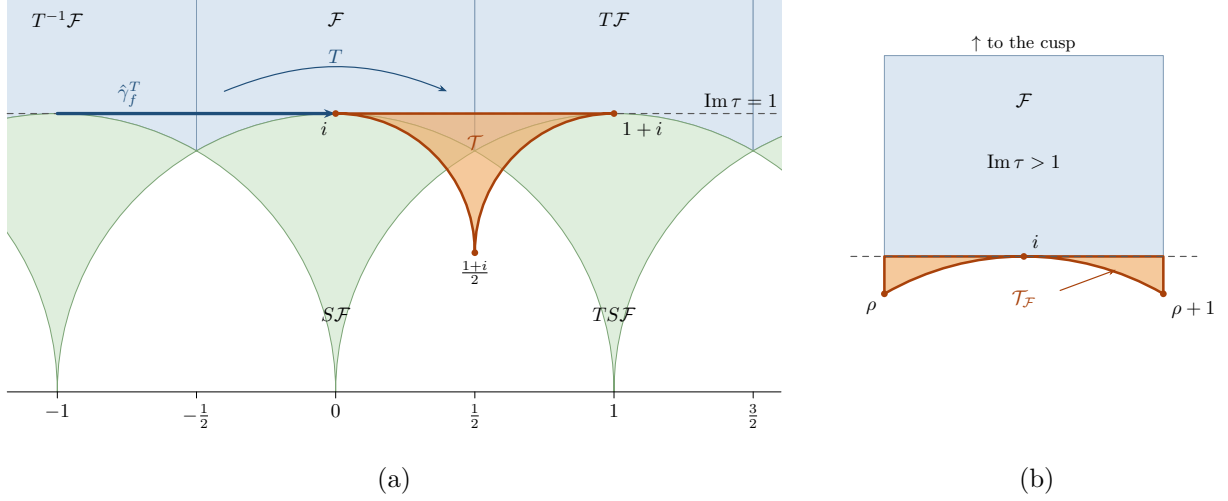


Figure 1: (a) The region $\mathcal{T}$ of Definition A.21, shaded, boundary in bold. The reference segment $\hat{\gamma}_f^T$ runs from $i - 1$ to i (arrowed): $\mathcal{T}$ is capped not by $\hat{\gamma}_f^T$ itself but by $T\hat{\gamma}_f^T$, the other two sides being $UT\hat{\gamma}_f^T$ and $U^2T\hat{\gamma}_f^T = S^{-1}\hat{\gamma}_f^T$, tangent at $\frac{1+i}{2}$; so $\partial\mathcal{T}$ is the U -orbit of $T\hat{\gamma}_f^T$ and $\mathcal{T}$ is U -invariant. It lies in no single copy of the fundamental domain, being cut by $\operatorname{Re} \tau = \frac{1}{2}$ and by $|\tau| = 1$, $|\tau - 1| = 1$, and so meets $\mathcal{F}$, $T\mathcal{F}$, $S\mathcal{F}$ and $TS\mathcal{F}$ alike. (b) All of $\mathcal{T}$ carried into $\mathcal{F}$: by (115) the shaded $\mathcal{T}_{\mathcal{F}}$ is the sliver $\{\tau \in \mathcal{F} : \operatorname{Im} \tau \leq 1\}$, of area $\frac{\pi}{3} - 1 = \frac{1}{3}\operatorname{area}(\mathcal{T})$, a fraction $1 - 3/\pi \approx 0.045$ of $\mathcal{F}$. Poles are not excluded there; they are the ones carrying the correction Q_f of Lemma A.26. Note $\mathcal{T}_{\mathcal{F}}$ is twice the part of $\mathcal{T}$ visible inside $\mathcal{F}$ in (a), the T -, S - and TS -parts folding onto $\operatorname{Re} \tau < 0$.

Definition A.22. *Define*

$$F_k^\circ := \left\{ f \in F_k : f^{-1}(\infty) \cap \mathcal{T} = \emptyset \text{ and } \int_{\hat{\gamma}_f^T} f(\tau) d\tau = 0 \right\}. \quad (114)$$

The second condition is the vanishing of the constant Fourier mode of f in the strip containing $\hat{\gamma}_f^T$, and reduces to $c_f(0) = 0$ when f has no poles above that segment.

Definition A.23. *For $f \in F_k^\circ$, let r_f be given by Definition 3.7, with $L^*(f, s)$ evaluated in the reference class of Definition A.3.*

Our definitions of $\mathcal{T}$ and F_k° are designed so that the proof of Lemma 3.11, which used $f \in S_k^!$ in exactly two places, extends to $f \in F_k^\circ$:

Lemma A.24. *For $f \in F_k^\circ$, the period polynomial $r_f \in W \subset V_n$.*

Proof. Lemma 3.11 invokes $f \in S_k^!$ only through $\hat{C}_T = 0$ and $\hat{C}_{\pm I} = 0$, and the two conditions of (114) supply exactly these.

For $\hat{C}_T$, the computation in the proof of Lemma 3.9 gives $\hat{C}_T = -(2\pi i)^{n+1} A_0 \int_{\hat{\gamma}_f^T} f(\tau) d\tau$, which vanishes by the second condition of (114).

For $\hat{C}_{\pm I}$, n is even, so $-I$ acts trivially on V_n and the coboundary drops out, leaving $\hat{C}_{-I} = C_{-I}$. Since $U^3 = -I$, the relation at $-I$ is the $(1 + U + U^2)$ loop, and by Definition A.21 its geometric realisation is $\partial\mathcal{T}$: thus C_{-I} is the integral of $f(\tau)(X - \tau Y)^n$ around the closed contour $\partial\mathcal{T}$. This is the step the region $\mathcal{T}$ exists to serve. The integrand is holomorphic wherever f is, and the first condition of (114) says precisely that f has no pole on or within $\mathcal{T}$, so Cauchy's theorem gives

$C_{-I} = 0$. Were even one pole of f to lie in $\mathcal{T}$, the loop would enclose it and C_{-I} would collect its residue instead; that residue is the defect computed in Lemma A.26 below, and it is what obstructs $r_f \in W$ for general $f \in F_k$.

The remaining steps are formal consequences of the cocycle law and apply verbatim. $\square$

The two conditions in (114) are not independent. An orbit meets $\mathcal{T}$ exactly when its representative p in the standard fundamental domain has $\text{Im } p \leq 1$: for such a p , $|p| \geq 1$ gives $|p - \frac{i}{2}|^2 = |p|^2 - \text{Im } p + \frac{1}{4} \geq \frac{1}{4}$ and likewise at the cusp 1, so p lies outside both circles, while $\text{Im } p \geq \frac{\sqrt{3}}{2} > \frac{1}{2}$ holds throughout the standard fundamental domain; hence p lies in $\mathcal{T}$ after a translation. Conversely no point of $\mathcal{T}$ can be raised above the contour, since $\text{Im}(\gamma\tau) = \text{Im } \tau / |c\tau + d|^2 > 1$ would need $|c\tau + d|^2 < \text{Im } \tau \leq 1$: for $c = 0$ this asks $|d| < 1$; for $|c| \geq 2$ it fails because $|c\tau + d|^2 \geq c^2(\text{Im } \tau)^2 \geq 1$ once $\text{Im } \tau \geq \frac{1}{2}$; and for $|c| = 1$ it asks exactly that τ lie inside $|\tau - m - \frac{i}{2}| < \frac{1}{2}$ for some $m \in \mathbb{Z}$, which is empty on $0 \leq \text{Re } \tau \leq 1$ unless $m \in \{0, 1\}$. The last two bounds of (113) are therefore not ad hoc: $\text{Im } \tau \geq \frac{1}{2}$ disposes of $|c| \geq 2$, and the two excluded disks are the $|c| = 1$ isometric circles. So $\mathcal{T}$ is precisely the part of the strip $0 \leq \text{Re } \tau \leq 1$ at or below the contour that no $\gamma \in \text{SL}_2(\mathbb{Z})$ can lift above it, and

$$\mathcal{T}_{\mathcal{F}} := \text{SL}_2(\mathbb{Z}) \cdot \mathcal{T} \cap \mathcal{F} = \{\tau \in \mathcal{F} : \text{Im } \tau \leq 1\}. \quad (115)$$

The first condition therefore says that every pole of f lies above $\hat{\gamma}_f^T$. This holds for a fraction $3/\pi \approx 0.955$ of pole positions in hyperbolic measure, the region $\text{Im } \tau > 1$ having area 1 against $\pi/3$ for the fundamental domain. But it also separates $\hat{\gamma}_f^T$ from the cusp, so the strip constant is $c_f(0)$ less $2\pi i$ times the residues in one period strip, and the second condition is then a further linear condition rather than a consequence of $c_f(0) = 0$. A form vanishing at the cusp whose poles form a single orbit of simple poles has strip constant exactly $2\pi i r_{f,p}^*(1)$, and so lies in F_k° only if that residue vanishes; membership requires $c_f(0)$ tuned against the residues, as for $E_{12}(j-j_1)/(j-j(2i))$ with j_1 chosen suitably [9]. Figure 1b draws $\mathcal{T}_{\mathcal{F}}$, whose area is $\int_{-1/2}^{1/2} ((1-x^2)^{-1/2} - 1) dx = \frac{\pi}{3} - 1$.

Poles inside $\mathcal{T}$ do not destroy this structure; they shift r_f by an explicit polynomial. As the sides of $\mathcal{T}$ are U -transports of one path, the enclosed pole set is a union of U -orbits $\{p, Up, U^2p\}$, and each enclosed pole contributes its residue.

Definition A.25. Let $f \in F_k$ have a pole of order m_p at each $p \in f^{-1}(\infty) \cap \mathcal{T}^\circ$, with projection weights $r_{f,p}^*(\alpha)$ (Definition A.6). The polar defect polynomial of f is

$$Q_f(X, Y) := (2\pi i)^{n+2} \sum_{p \in f^{-1}(\infty) \cap \mathcal{T}^\circ} \sum_{\alpha=1}^{m_p} r_{f,p}^*(\alpha) B_\alpha(p; X, Y), \quad (116)$$

$$B_\alpha(p; X, Y) := \binom{n}{\alpha-1} (-Y)^{\alpha-1} (X - pY)^{n-\alpha+1}. \quad (117)$$

Simple poles contribute the $\alpha = 1$ term alone, $r_{f,p}^*(1)(X - pY)^n$.

Lemma A.26. Let $f \in F_k$ satisfy $\int_{\hat{\gamma}_f^T} f(\tau) d\tau = 0$. Then

$$r_f|_{(1+S)} = 0 \quad \text{and} \quad r_f|_{(1+U+U^2)} = -Q_f. \quad (118)$$

In particular $r_f \in W$ if and only if $Q_f = 0$, and Lemma A.24 is the case $f^{-1}(\infty) \cap \mathcal{T}^\circ = \emptyset$.

Proof. Both relations are values of the cocycle of Lemma 3.9 on the two relators of $\mathrm{SL}_2(\mathbb{Z})$, evaluated on the loops their words induce; the loops, not the group elements, carry the content. For the first, $\hat{C}_{S^2} = \hat{C}_S|_{(1+S)} = r_f|_{(1+S)}$; as n is even the coboundary drops out, so this is the integral of $(2\pi i)^{n+1} f(\tau)(X - \tau Y)^n$ over the S^2 -loop, the concatenation of $\hat{\gamma}_f^S$ with its S -image. At $\delta = 0$ that loop is degenerate. For $\delta > 0$ it retraces itself: the reference S -segment runs up the imaginary axis from $i/(1+\delta)$ to $i(1+\delta)$, and S carries iy to i/y , mapping the segment onto itself with reversed orientation, so the two legs cancel. Definition A.3 admits no pole in $1 < \mathrm{Im} \tau \leq 1+\delta$, and applying S excludes them from $1/(1+\delta) \leq \mathrm{Im} \tau < 1$ as well, so the only pole that can meet the segment is $\tau = i$. There S acts with derivative -1 , preserving the symmetric excision, and the two principal values cancel with the rest. Either way $r_f|_{(1+S)} = 0$ [9]. For the second, the hypothesis $\int_{\hat{\gamma}_f^T} f d\tau = 0$ gives $\hat{C}_T = 0$ as in the proof of Lemma A.24, hence $\hat{C}_U = r_f$ and $r_f|_{(1+U+U^2)} = \hat{C}_{U^3} = C_{-I}$, the integral of the same 1-form over the $(TS)^3$ -loop: the three U -transports of the U -path traverse $\partial\mathcal{T}$ once clockwise, with a common integrand by modular covariance. The residue theorem then gives $C_{-I} = -(2\pi i)^{n+1} \cdot 2\pi i \sum_{p \in \mathcal{T}^\circ} \mathrm{Res}_{\tau=p} [f(\tau)(X - \tau Y)^n]$, and since $\frac{1}{(\alpha-1)!} \partial_\tau^{\alpha-1} (X - \tau Y)^n = B_\alpha(\tau; X, Y)$, that residue is $\sum_{\alpha \leq m_p} r_{f,p}^*(\alpha) B_\alpha(p; X, Y)$; hence $C_{-I} = -Q_f$. $\square$

A pole of f can meet $\partial\mathcal{T}$ only at the elliptic point i . The sides of $\mathcal{T}$ are Γ -translates of $\hat{\gamma}_f^T$ and the poles of f are Γ -invariant, so a pole on $\partial\mathcal{T}$ would lie on $\hat{\gamma}_f^T$ itself, which Definition A.3 excludes by lifting the base-point to $i(1+\delta)$; when $\delta > 0$ the S -segment no longer degenerates and $\partial\mathcal{T}$ acquires three further sides translating $\hat{\gamma}_f^S$, so (113) is the $\delta = 0$ form, and i is taken there by principal value.

The defect is removed by subtracting from f reference forms carrying the same poles. Write $\mathrm{SL}_2(\mathbb{Z}) \cdot p := \{\gamma \cdot p : \gamma \in \mathrm{SL}_2(\mathbb{Z})\}$ for the orbit of $p \in \mathbb{H}$.

Write $\Gamma_p \leq \mathrm{SL}_2(\mathbb{Z})$ for the stabiliser of p , and let $A_p := \{\alpha : 1 \leq \alpha \leq m, B_\alpha(p; \cdot, \cdot) \text{ has nonzero } \Gamma_p\text{-average}\}$. At a non-elliptic p one has $\Gamma_p = \{\pm I\}$ and A_p is all of $\{1, \dots, m\}$; at $\tau = i$ and $\tau = e^{i\pi/3}$ it is a proper subset, namely the $\alpha \equiv k/2 \pmod{2}$ and the $\alpha \equiv -k \pmod{3}$ respectively [9], which are exactly the pole orders the valence formula permits there.

Definition A.27. A reference family at p of depth m is a set $\{\Psi_p^{(\alpha)}\}_{\alpha \in A_p}$ of weight- k meromorphic modular forms, each holomorphic on $\mathbb{H} \setminus \mathrm{SL}_2(\mathbb{Z}) \cdot p$ with constant Fourier coefficient zero, such that

$$\mathrm{Res}_{\tau=p} [\Psi_p^{(\alpha)}(\tau)(X - \tau Y)^n d\tau] = \sum_{\beta \in A_p, \beta \leq \alpha} \lambda_{\alpha\beta} B_\beta(p; X, Y), \quad \lambda_{\alpha\alpha} \neq 0, \quad (119)$$

that is, triangular in the B_β with nonvanishing diagonal.

Two constructions. Taking $\Psi_p^{(\alpha)} := g_\alpha / (j - j(p))^\alpha$ with $g_\alpha \in M_k$ not vanishing at p gives a reference family whenever such g_α exist, since a pole of exact order α contributes $\lambda_{\alpha\alpha} = r_{\Psi(\alpha),p}^*(\alpha) \neq 0$ and nothing beyond. Alternatively the Poincaré series $\Psi_{\mathrm{SL}_2(\mathbb{Z})}^{\alpha-1, n-\alpha+1}(\cdot, p)$ of [15, Def. 3.1] form one at every p : they have weight k , are holomorphic off $\mathrm{SL}_2(\mathbb{Z}) \cdot p$ with poles there of order at most α , and vanish at the cusp [15, Prop. 3.3], while by [15, Cor. 3.11] their residue is a multiple of $(X - \bar{p}Y)^{\alpha-1} (X - pY)^{n-\alpha+1}$, which on expanding $X - \bar{p}Y = (X - pY) - (p - \bar{p})(-Y)$ is triangular in the B_β with diagonal entry proportional to $(p - \bar{p})^{\alpha-1}$. That frame agrees with ours, their (X, Y) being dual to $(1, -\tau)$ so that their $X - \tau Y$ is the class of du on $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$; the multiple is a local character sum $\kappa^{\alpha-1, n-\alpha+1}$, equal by [15, Rem. 3.12] to $|\Gamma_p|$ for $\alpha \in A_p$ and to zero otherwise.

Theorem A.28. Let $f \in F_k$ have $c_f(0) = 0$, with poles in $\mathbb{H}$ on the orbits $\mathrm{SL}_2(\mathbb{Z}) \cdot p_1, \dots, \mathrm{SL}_2(\mathbb{Z}) \cdot p_N$ of orders $m_1, \dots, m_N$, elliptic or not; a pole at the cusp is allowed. Fix a reference family $\Psi_{p_a}^{(\alpha)}$ at

each p_a of depth m_a , and evaluate every period polynomial below in the reference class of f . This is available to the $\Psi_{p_a}^{(\alpha)}$ as well: their poles lie among those of f , so Definition A.3 admits none of them in $1 < \text{Im } \tau \leq 1 + \delta$, and the same contour serves every integrand. There are unique $c_{a,\alpha} \in \mathbb{C}$, for $1 \leq a \leq N$ and $\alpha \in A_{p_a}$, such that $f - \sum_{a,\alpha} c_{a,\alpha} \Psi_{p_a}^{(\alpha)}$ is holomorphic on $\mathbb{H}$, and then

$$\tilde{r}_f := r_f - \sum_{a=1}^N \sum_{\alpha \in A_{p_a}} c_{a,\alpha} r_{\Psi_{p_a}^{(\alpha)}} \in W. \quad (120)$$

Both r_f and each $r_{\Psi_{p_a}^{(\alpha)}}$ are given by Definition 3.7, so $\tilde{r}_f$ is assembled from the values $L^*(\cdot, \ell + 1)$, $0 \leq \ell \leq n$, of the finite-contour L -function alone, and the $c_{a,\alpha}$ are determined by the principal parts of f .

Proof. Fix a and write $p = p_a$. The residue of $f(\tau)(X - \tau Y)^n d\tau$ at p is $\sum_{\alpha \leq m_a} r_{f,p}^*(\alpha) B_\alpha(p)$ as in (116). That residue is Γ_p -invariant: the 1-form is $\text{SL}_2(\mathbb{Z})$ -invariant and Γ_p fixes p , so averaging over Γ_p leaves it unchanged. Hence it lies in the span of $\{B_\beta(p) : \beta \in A_p\}$, which is the image of that average. The residues (119) of the reference family are triangular in the same B_β , $\beta \in A_p$, with nonvanishing diagonal, hence a basis of that span; so the $c_{a,\alpha}$ matching the two exist and are unique. At a non-elliptic p this is the whole span and the statement is the evident one; at $\tau = i$ or $e^{i\pi/3}$ the individual Laurent coefficients $r_{f,p}^*(\alpha)$ with $\alpha \notin A_p$ need not vanish, but their contributions cancel in the invariant combination.

Since $\Psi_{p_a}^{(\alpha)}$ has no pole off $\text{SL}_2(\mathbb{Z}) \cdot p_a$, the orbits do not interfere and the choices are independent in a . Matching the residue at p_a matches the entire principal part there, and covariance $r_{g,\gamma p}^*(\alpha) = (cp + d)^n r_{g,p}^*(\alpha)$ propagates this along the orbit, so the difference has no pole in $\mathbb{H}$ and lies in $M_k^!$. Its constant Fourier coefficient is $c_f(0) = 0$, so it lies in $S_k^!$, where Lemma 3.11 gives a period polynomial in W ; r is linear. $\square$

The hypothesis $c_f(0) = 0$ costs nothing. For arbitrary $f \in F_k$ the form $f - c_f(0)E_k$ has the same poles in $\mathbb{H}$ and vanishing constant term, so Theorem A.28 applies to it, and

$$\hat{r}_f := \tilde{r}_{f - c_f(0)E_k} + c_f(0) r_{E_k} \quad (121)$$

is defined for every $f \in F_k$. It satisfies $\hat{r}_f|_{(1+S)} = 0$, since both terms do, and $\hat{r}_f|_{(1+U+U^2)} = c_f(0) r_{E_k}|_{(1+U+U^2)}$. So $\hat{r}_f$ lies in W exactly when $c_f(0) = 0$, and otherwise in $W \oplus \mathbb{C} r_{E_k}$; this is the usual Eisenstein correction, already present for holomorphic $f \in M_k \setminus S_k$, and not a feature of the poles.

Corollary A.29. *Let $k = 12$ and $f = \Delta/(j - j_0)$ with $j_0 \notin \{0, 1728\}$, and take the reference family $\Psi_p^{(1)} := E_{12}/(j - j_0)$ at a pole p . Then $N = 1$, $m_1 = 1$,*

$$c_{1,1} = \frac{\Delta(p)}{E_{12}(p)} = \frac{691}{691j_0 - 432000}, \quad \tilde{r}_f = -c_{1,1} r_\Delta. \quad (122)$$

In particular $c_{1,1}$ is rational whenever j_0 is, with no assumption of complex multiplication, and $\tilde{r}_f$ is then a rational multiple of the period polynomial of Δ .

Proof. $E_{12}/(j - j_0)$ is holomorphic off $\text{SL}_2(\mathbb{Z}) \cdot p$, has a simple pole there with nonzero residue, and vanishes at the cusp, so it is a reference family of depth 1. From $E_4^3 = j\Delta$ and $E_6^2 = (j - 1728)\Delta$ one gets $E_{12} = \Delta(691j - 432000)/691$, so Δ/E_{12} takes the stated value at p , and substituting gives $f - c_{1,1}\Psi_p^{(1)} = -c_{1,1}\Delta$ identically. Apply Theorem A.28 and linearity. $\square$

Corollary A.30. *Let $d > 4$ have class number one, let α_d be the associated CM point, and set $j_d := j(\alpha_d) \in \mathbb{Z}$. Write $D := \frac{1}{2\pi i} \frac{d}{d\tau}$, and let H_d be the Hilbert class polynomial, whose roots are the singular moduli of discriminant d [18]. Then*

$$\Lambda_d := D \log H_d(j) = \frac{-E_4^2 E_6}{\Delta} \frac{H'_d(j)}{H_d(j)} = \frac{-E_4^2 E_6}{\Delta(j - j_d)} \quad (123)$$

is a weight-two form with a simple pole of residue $1/2\pi i$ at every CM point of discriminant d , the last equality holding because $h(d) = 1$ makes $H_d(j) = j - j_d$. Moreover

$$f_d := \Delta E_6 \Lambda_d = \frac{-E_4^2 E_6^2}{j - j_d} \in F_{20} \quad (124)$$

has integral q -expansion, vanishes at the cusp, and no pole of f_d lies in $\mathcal{T}$; hence $\tilde{r}_{f_d} = r_{f_d} - c_{f_d} r_{E_{20}} \in W$ is determined, with no reference family and no choice. Explicitly, with ΔE_8 the normalised cusp form spanning S_{20} ,

$$\tilde{r}_{f_d} = \frac{-92207808}{174611j_d - 209520000} r_{\Delta E_8}. \quad (125)$$

Proof. $Dj = -E_4^2 E_6 / \Delta$ gives the second expression in (123), and $\Delta E_6 \Lambda_d$ reduces to (124); the coefficients are integral because $j_d \in \mathbb{Z}$ for class number one. Since $h(d) = 1$ the only reduced form of discriminant d has leading coefficient 1, so $\text{Im } \alpha_d = \frac{\sqrt{d}}{2} > 1$ for $d > 4$; by the criterion above, its orbit misses $\mathcal{T}$, and $Q_{f_d} = 0$. For (125), $E_8 = E_4^2$ and $E_6^2 = (j - 1728)\Delta$ give

$$f_d = \frac{-\Delta E_8 (j - 1728)}{j - j_d} = -\Delta E_8 - (j_d - 1728) \frac{\Delta E_8}{j - j_d}. \quad (126)$$

Running the computation of Corollary A.29 with (Δ, E_{12}) replaced by $(\Delta E_8, E_{20})$, where $E_{20} = \Delta E_8 (174611j - 209520000) / 174611$, the reference family $E_{20} / (j - j_d)$ gives $c = 174611 / (174611j_d - 209520000)$ and $\tilde{r}_{\Delta E_8 / (j - j_d)} = -c r_{\Delta E_8}$. Since $\Delta E_8 \in S_{20}$, linearity of L^* turns (126) into $\tilde{r}_{f_d} = [(j_d - 1728)c - 1] r_{\Delta E_8}$, and $209520000 - 1728 \cdot 174611 = -92207808$ gives (125). $\square$

These are the discriminants $d = 7, 8, 11, 19, 43, 67, 163$. For $d = 3$ and $d = 4$ the CM point is elliptic and $\text{Im } \alpha_d \leq 1$, so both the elliptic bookkeeping of Definition A.27 and a correction are needed; and for $h(d) > 1$ the reduced forms with leading coefficient $a > 1$ give CM points as low as $\text{Im} = \frac{\sqrt{3}}{2}$, whose orbits do meet $\mathcal{T}$.

Definition A.27 constrains the $\Psi_p^{(\alpha)}$ only through their poles, so it does not determine them, and $\tilde{r}_f$ inherits that freedom. The freedom is exactly measured. If $\{\Psi_p^{(\alpha)}\}$ and $\{\Psi'_p^{(\alpha)}\}$ are two reference families and $c_{a,\alpha}, c'_{a,\alpha}$ the coefficients they produce for the same f , then $\sum c_{a,\alpha} \Psi_{p_a}^{(\alpha)}$ and $\sum c'_{a,\alpha} \Psi_{p_a}'^{(\alpha)}$ have the same principal part as f at every pole, so their difference g is holomorphic on $\mathbb{H}$ with $c_g(0) = 0$, that is $g \in S_k^!$, and

$$\tilde{r}_f - \tilde{r}'_f = r_g \in W \quad (127)$$

by Lemma 3.11. The ambiguity is therefore not merely W -valued by accident: it is the set $r(S_k^!) \subseteq W$ of period polynomials of the pole-free forms, which is precisely what Definition A.27 cannot see.

That set is everything. Adding any $h \in S_k^!$ to each $\Psi_{p_a}^{(\alpha)}$ yields another reference family, since h is holomorphic with $c_h(0) = 0$ and so alters neither the pole locus nor any residue in (119); the matched coefficients are unchanged, and $\tilde{r}_f$ moves by r_g with $g = \sum c_{a,\alpha} h_\alpha$ arbitrary in $S_k^!$. And r is already onto W there, for two independent reasons. So long as the determinant pairing the $\dim S_k$

periods against the $\dim S_k$ quasi-periods is nonzero, which is the Legendre relation [7], the $2 \dim S_k$ period polynomials of the cusp forms and of their weakly holomorphic partners are independent modulo p_0 , and so span a complement to $\mathbb{C}p_0$ in W . The remaining Eisenstein direction p_0 is supplied separately, by the Bol image $\partial^{k-1}M_{2-k}^!$: it lies in $S_k^!$, since ∂^{k-1} annihilates the constant Fourier mode, and its period polynomials are coboundaries, hence multiples of p_0 — the only coboundary compatible with $\hat{C}_T = 0$, as Y^n spans the T -invariants of V_n [9]. So Theorem A.28 determines nothing about $\tilde{r}_f$ beyond its membership in W , and pinning it down requires restricting Definition A.27.

Two restrictions suggest themselves, and they disagree. One may fix the reference forms once and for all, taking $\Psi_p^{(\alpha)}$ to be the Poincaré series of [15, Def. 3.1], which are canonically defined; no freedom then remains, at the price of resting on a normalisation that [15, Rem. 5.17] notes is not rational. Alternatively one may define $\tilde{r}_f$ by continuation in the pole location. For f with no pole in $\mathcal{T}$, Lemma A.24 gives $r_f \in W$ with no reference form at all; moving a pole into $\mathcal{T}$ crosses a side of $\partial\mathcal{T}$, which is a Γ -translate of $\hat{\gamma}_f^T$, so r_f jumps by the crossing of Lemma A.4, and continuing through that jump determines $\tilde{r}_f$ up to which side was crossed. The residual freedom is then generated by the windings about the points Sp , one per enclosed orbit, and is a proper subspace of W : winding about any other point leaves W altogether and so is not an admissible correction [9].

The two can be compared exactly. Take f with a single orbit of simple poles, none in $\mathcal{T}$, so that the continuation prescription reduces to $r_f - c_f r_{E_k}$ with no reference form. Since $\Psi_p^{(1)}$ also vanishes at the cusp, its strip constant is $-2\pi i$ times its residue, and the residue-matching coefficient c therefore satisfies $c c_\Psi = c_f$; subtracting,

$$(r_f - c_f r_{E_k}) - (r_f - c r_{\Psi_p^{(1)}}) = c (r_{\Psi_p^{(1)}} - c_\Psi r_{E_k}). \quad (128)$$

The right side is c times the corrected period polynomial of the reference form itself. The two prescriptions agree precisely when that vanishes, and since $r(S_k^!) = W$ a family with $r_\Psi - c_\Psi r_{E_k} = 0$ can always be arranged by adding a suitable element of $S_k^!$; for $\Psi_p^{(1)} = E_k/(j - j(p))$ it does not vanish [9]. Which normalisation is the right one we leave open, noting only that the continuation prescription agrees by construction with the unambiguous value on the complement of $\mathcal{T}$, where no reference form is called for.

Remark A.31. The polar data used above is not new. Brown and Fonseca [15] compute the residue of $f(\tau)(X - \tau Y)^n d\tau$ at a pole w , in their Lemma 5.10, as $\sum_j \binom{n}{j} c_{-j-1}(w)(X - wY)^{n-j}(-Y)^j$ with $c_{-j-1}(w)$ the Laurent coefficients of f there; under $j = \alpha - 1$ this is Definition A.25 term by term. The reference forms $\Psi_p^{(\alpha)}$ of (119) are theirs, as are the residue evaluation used in the proof of Theorem A.28, the local character sums fixing A_p , and the identification of $E_k/(j - j(p))$ in the simple-pole case. What is added here is the L -function: r_f and the $r_{\Psi_p^{(\alpha)}}$ are values of the finite-contour L^* , and the region $\mathcal{T}$, which decides by Lemma A.24 whether any correction is needed at all, presupposes a contour and has no counterpart in their setting.

The dependence of $\tilde{r}_f$, and hence of $\hat{r}_f$ in (121), on the choice of reference family is (127). For adjacent work see [7] for poles at the cusp, [21] for meromorphic forms with restricted poles, and [22] for regularised inner products. The forms $f_{k,P}$ of [20], going back to [17], have weight $2k$ and poles of order k at CM points, and supply test cases for (120) of arbitrary order.

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